

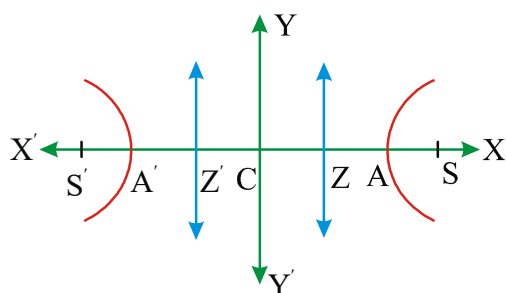
HYPERBOLA



SYNOPSIS

Introduction :

- A Conic is said to be a hyperbola if its eccentricity is greater than one.
- If $S = ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$ represents a hyperbola then $h^2 - ab > 0$ and $\Delta \neq 0$
- If $S = ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$ represents a rectangular hyperbola then $h^2 - ab > 0$, $\Delta \neq 0$ and $a + b = 0$
- The centre of the hyperbola $S = 0$ is obtained by solving the equations of axes $\frac{\partial S}{\partial x} = 0, \frac{\partial S}{\partial y} = 0$
- **Hyperbola:** Equation of the hyperbola in standard form is $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$.



- Hyperbola is symmetric about both the axes.
- Hyperbola does not pass through the origin and does not meet the Y - axis.
- Hyperbola curve does not exist between the lines $x = -a$ and $x = a$.

Notation: $S = \frac{x^2}{a^2} - \frac{y^2}{b^2} - 1$, $S_1 = \frac{xx_1}{a^2} - \frac{yy_1}{b^2} - 1$,

$S_{11} = \frac{x_1^2}{a^2} - \frac{y_1^2}{b^2} - 1$, $S_{12} = \frac{x_1x_2}{a^2} - \frac{y_1y_2}{b^2} - 1$.

- **Rectangular hyperbola (or) Equilateral hyperbola :** In a hyperbola if the length of the transverse axis ($2a$) is equal to the length of the conjugate axis ($2b$), then the hyperbola is called a rectangular hyperbola or an equilateral hyperbola.
- The eccentricity of a rectangular hyperbola is $\sqrt{2}$.

- **Conjugate hyperbola :** The hyperbola whose transverse and conjugate axes are respectively the conjugate and transverse axes of a given hyperbola is called the conjugate hyperbola of the given hyperbola.

- The equation of the hyperbola conjugate to $S = 0$ is $S^1 = \frac{x^2}{a^2} - \frac{y^2}{b^2} + 1 = 0$.

- If e_1 and e_2 be the eccentricities of the hyperbola and its conjugate hyperbola then $\frac{1}{e_1^2} + \frac{1}{e_2^2} = 1$.

- **Parametric Equations:** The parametric equations of a hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ are $x = a \sec \theta$ and $y = b \tan \theta$ where θ is parameter ($0 \leq \theta < 2\pi$). A point on the hyperbola is represented by $(a \sec \theta, b \tan \theta)$

Note : A point on the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ can also be represented by $(a \cosh \theta, b \sinh \theta)$. The equations $x = a \cosh \theta, y = b \sinh \theta$ are called the

parametric equations of the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$

- **Auxiliary Circle :** The circle described on the transverse axis of a hyperbola as diameter is called the auxiliary circle of the hyperbola
- The locus of the feet of the perpendiculars from foci on any tangent of the hyperbola is a circle concentric with the hyperbola. This circle is called Auxiliary circle of the hyperbola.
- The equation of auxiliary circle of $S=0$ is $x^2 + y^2 = a^2$
- **Director Circle:** The locus of point of intersection of perpendicular tangents to a hyperbola is a circle concentric with hyperbola. This circle is called director circle of the hyperbola
- The equation to the director circle of $S = 0$ is $x^2 + y^2 = a^2 - b^2$

VARIOUS FORMS OF HYPERBOLA

	HYPERBOLA	CONJUGATE HYPERBOLA
Standard Equation	$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$	$\frac{x^2}{a^2} - \frac{y^2}{b^2} = -1$
Centre	(0,0)	(0,0)
Eccentricity	$e = \frac{\sqrt{a^2 + b^2}}{a}$ (or) $b^2 = a^2(e^2 - 1)$	$e = \frac{\sqrt{a^2 + b^2}}{b}$ (or) $a^2 = b^2(e^2 - 1)$
Foci	($\pm ae$, 0)	(0, $\pm be$)
Vertices	($\pm a$, 0)	(0, $\pm b$)
Length of the latusrectum	$\frac{2b^2}{a}$	$\frac{2a^2}{b}$
Length of the transverse axis	2a	2b
Length of the conjugate axis	2b	2a
Equations of directrices	$x = \pm \frac{a}{e}$	$y = \pm \frac{b}{e}$
Equations of latusrecta	$x = \pm ae$	$y = \pm be$
Equations of the transverse axis	$y = 0$	$x = 0$
Equations of the conjugate axis	$x = 0$	$y = 0$
Equations of the director circle	$x^2 + y^2 = a^2 - b^2$	$x^2 + y^2 = b^2 - a^2$
Equations of the auxiliary circle	$x^2 + y^2 = a^2$	$x^2 + y^2 = b^2$
if P is any point on the hyperbola and S, S' are foci then	$ SP - S'P = 2a$	$ SP - S'P = 2b$
Ends of latusrectum	$(\pm ae, \pm \frac{b^2}{a})$	$(\pm \frac{a^2}{b}, \pm be)$

	HYPERBOLA	CONJUGATE HYPERBOLA
Standard Equation with centre (h,k)	$\frac{(x-h)^2}{a^2} - \frac{(y-k)^2}{b^2} = 1$	$\frac{(x-h)^2}{a^2} - \frac{(y-k)^2}{b^2} = -1$
Centre	(h, k)	(h, k)
Eccentricity	$e = \frac{\sqrt{a^2 + b^2}}{a}$	$e = \frac{\sqrt{a^2 + b^2}}{b}$
Foci	(h ± ae, k)	(h, k ± be)
Vertices	(h ± a, k)	(h, k ± b)
Length of the latusrectum	$\frac{2b^2}{a}$	$\frac{2a^2}{b}$
Length of the transverse axis	2a	2b
Length of the conjugate axis	2b	2a
Equations of directrices	$x - h = \pm \frac{a}{e}$	$y - k = \pm \frac{b}{e}$
Equations of latusrecta	$x - h = \pm ae$	$y - k = \pm be$
Equations of the transverse axis	$y - k = 0$	$x - h = 0$
Equations of the conjugate axis	$x - h = 0$	$y - k = 0$
Equations of the director circle	$(x - h)^2 + (y - k)^2 = a^2 - b^2$	$(x - h)^2 + (y - k)^2 = b^2 - a^2$
Equations of the auxiliary circle	$(x - h)^2 + (y - k)^2 = a^2$	$(x - h)^2 + (y - k)^2 = b^2$
if P is any point on the hyperbola and s, s' are foci then	$ SP - S'P = 2a$	$ SP - S'P = 2b$
Ends of latusrectum	$\left(h \pm ae, k \pm \frac{b^2}{a}\right)$	$\left(h \pm \frac{a^2}{b}, k \pm be\right)$

- **Position of a Point with respect to a hyperbola:** Let $P(x_1, y_1)$ be a point and

$$S = \frac{x^2}{a^2} - \frac{y^2}{b^2} - 1 = 0 \text{ be a hyperbola then}$$

- (i) p lies on the hyperbola iff $S_{11} = 0$.
 (ii) p lies inside (i.e. in the region not containing the centre) the hyperbola iff $S_{11} > 0$.
 (iii) p lies outside the hyperbola iff $S_{11} < 0$.

W.E-1 : If vertices and foci of a hyperbola are $(0, \pm 5)$ and $(0, \pm 8)$ respectively, then equation of hyperbola is

Sol: Since, vertices $(0, \pm 5)$ and foci $(0, \pm 8)$ lies on y-axis. Hence the equation of hyperbola will be of

$$\text{the form } -\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1 \quad \dots (i)$$

$$\therefore \text{vertices } (0, \pm b) = (0, \pm 5) \quad \text{and} \quad \text{foci } (0, \pm be) = (0, \pm 8)$$

$$b = 5 \text{ and } be = 8$$

$$a^2 = b^2(e^2 - 1) = b^2e^2 - b^2 = 64 - 25 = 39$$

On putting values of a^2 and b^2 in Eq. (i), we get

$$-\frac{x^2}{39} + \frac{y^2}{25} = 1$$

W.E-2 : The vertices of the hyperbola

$$9x^2 - 16y^2 - 36x + 96y - 252 = 0 \text{ are}$$

Sol: The equation can be rewritten as

$$9(x^2 - 4x + 4 - 4) - 16(y^2 - 6y + 9 - 9) = 252$$

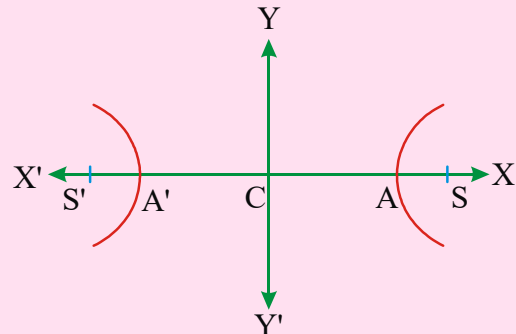
$$9(x - 2)^2 - 16(y - 3)^2 = 252 + 36 - 144 = 144$$

$$\Rightarrow \frac{(x - 2)^2}{16} - \frac{(y - 3)^2}{9} = 1$$

$$\text{Therefore, the vertices are } (h \pm a, b) = (2 \pm 4, 3)$$

$$\therefore \text{vertices are } (6, 3), (-2, 3)$$

W.E-3: The length of transverse axis of the hyperbola is 14. If the vertex bisects the distance between centre and focus then its equation is (in standard form)



$$\text{Sol. } 2a = 14 \Rightarrow a = 7$$

$$CA = AS \Rightarrow CA = CS - CA$$

$$\Rightarrow a = ae - a \Rightarrow e = 2$$

$$b^2 = a^2(e^2 - 1) = 147$$

$$\text{Equation of hyperbola is } \frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$$

$$\Rightarrow \frac{x^2}{49} - \frac{y^2}{147} = 1$$

W.E-4: Equations of the auxiliary circle and director circle of the hyperbola

$$9x^2 - 16y^2 + 72x - 32y - 16 = 0 \text{ are}$$

Sol: Given equation of hyperbola is

$$9x^2 - 16y^2 + 72x - 32y - 16 = 0$$

$$\Rightarrow 9(x + 4)^2 - 16(y + 1)^2 = 144$$

$$\Rightarrow \frac{(x + 4)^2}{16} - \frac{(y + 1)^2}{9} = 1$$

Equation of auxiliary circle is

$$(x - h)^2 + (y - k)^2 = a^2$$

$$\Rightarrow (x + 4)^2 + (y + 1)^2 = 16$$

$$\Rightarrow x^2 + y^2 + 8x + 2y + 1 = 0$$

Equation of director circle is

$$(x - h)^2 + (y - k)^2 = a^2 - b^2$$

$$\Rightarrow (x + 4)^2 + (y + 1)^2 = 16 - 9$$

$$\Rightarrow x^2 + y^2 + 8x + 2y + 10 = 0$$

W.E-5: If the foci of the ellipse $\frac{x^2}{16} + \frac{y^2}{b^2} = 1$ and

the hyperbola $\frac{x^2}{144} - \frac{y^2}{81} = \frac{1}{25}$ coincide, then

the value of b^2 is

Sol: For a hyperbola

$$e^2 = 1 + \frac{b^2}{a^2} = 1 + \frac{81}{144} = \frac{225}{144} \therefore e = \frac{15}{12} = \frac{5}{4}$$

also, $a^2 = \frac{144}{25}$ hence, the foci are

$$(\pm ae, 0) = \left(\pm \frac{12}{5} \cdot \frac{5}{4}, 0 \right) = (\pm 3, 0)$$

Now, for an ellipse, $ae = 3$ or $a^2 e^2 = 9$

Now, $b^2 = a^2 (1 - e^2)$

$$\Rightarrow b^2 = a^2 - a^2 e^2 = 16 - 9 = 7$$

- **Tangent to Hyperbola:** The equation of the tangent at $p(x_1, y_1)$ is $S_1 = 0$.
- The equation of the tangent at " θ " is $\frac{x}{a} \sec \theta - \frac{y}{b} \tan \theta = 1$.
- The condition for a straight line $y = mx + c$ to be a tangent to the hyperbola $S = 0$ is $c^2 = a^2 m^2 - b^2$ and the point of contact is $\left(\frac{-a^2 m}{c}, \frac{-b^2}{c} \right)$.
- The equation of the tangent to the hyperbola $S = 0$ having slope m is $y = mx \pm \sqrt{a^2 m^2 - b^2}$, $m < \frac{-b}{a}$, $m > \frac{b}{a}$
- The condition that the line $lx + my + n = 0$ may be a tangent to the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ is $a^2 l^2 - b^2 m^2 = n^2$ and the point of contact is $\left(\frac{-a^2 l}{n}, \frac{b^2 m}{n} \right)$.
- Two tangents can be drawn to a hyperbola from an external point.

- If m_1, m_2 are the slopes of the tangents through $P(x_1, y_1)$ to the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ then

$$m_1 + m_2 = \frac{2x_1 y_1}{x_1^2 - a^2}, \quad m_1 m_2 = \frac{y_1^2 + b^2}{x_1^2 - a^2}.$$

- The product of the perpendiculars from foci to any tangent to the hyperbola $S = 0$ is b^2 .
- If θ is the angle between the tangents from (x_1, y_1) to the hyperbola $S = 0$ then

$$\tan \theta = \frac{2ab\sqrt{-S_{11}}}{\left| x_1^2 + y_1^2 - a^2 + b^2 \right|}.$$

- **Normal to Hyperbola:** The equation of the normal to the hyperbola $S = 0$ at (x_1, y_1) is

$$\frac{a^2 x}{x_1} + \frac{b^2 y}{y_1} = a^2 + b^2.$$

- The equation of the normal at θ is

$$\frac{ax}{\sec \theta} + \frac{by}{\tan \theta} = a^2 + b^2, \quad \left(\theta \neq 0, \frac{\pi}{2}, \pi \right)$$

- The condition that the line $lx + my + n = 0$ may be normal to the hyperbola $S = 0$ is

$$\frac{a^2}{l^2} - \frac{b^2}{m^2} = \frac{(a^2 + b^2)^2}{n^2}.$$

- The condition that the line $y = mx + c$ to be a normal to the hyperbola $S = 0$ is

$$c = \pm \frac{m(a^2 + b^2)}{a^2 - b^2}$$

- Equation of normal to the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$

$$\text{in slope form is } y = mx \pm \frac{m(a^2 + b^2)}{a^2 - b^2}$$

- The equation of common tangent to the hyperbola

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1 \text{ and } \frac{y^2}{a^2} - \frac{x^2}{b^2} = 1 \text{ are}$$

$$y = \pm x \pm \sqrt{a^2 - b^2}$$

$$\text{The length of common tangent is } \sqrt{2} \frac{(a^2 + b^2)}{\sqrt{a^2 - b^2}}$$

- **Co-normal points:** In general four normals can be drawn to a hyperbola from any point. If $\alpha, \beta, \gamma, \delta$ be the eccentric angles of these four co-normal points then $\alpha + \beta + \gamma + \delta$ is an odd multiple of π i.e., Sum of the eccentric angles of four co-normal points is an odd multiple of π .
- If α, β, γ are the eccentric angles of three co-normal points on the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$, the normals at which are concurrent, then $\sin(\alpha + \beta) + \sin(\beta + \gamma) + \sin(\gamma + \alpha) = 0$
- The equation of the chord joining the two points α and β on the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ is
- $$\frac{x}{a} \cos\left(\frac{\alpha - \beta}{2}\right) - \frac{y}{b} \sin\left(\frac{\alpha + \beta}{2}\right) = \cos\left(\frac{\alpha + \beta}{2}\right)$$
- If $P(x_1, y_1) = (a \sec \theta, b \tan \theta)$ is a point on $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ and S, S' are foci then.
- $$SP = |ex_1 - a| = |ae \sec \theta - a| \text{ and } S'P = |ex_1 + a| = |ae \sec \theta + a|$$
- If $P(x_1, y_1)$ is a point on $\frac{x^2}{a^2} - \frac{y^2}{b^2} + 1 = 0$ and S, S' are foci then
- $$SP = |ey_1 - b|; S'P = |ey_1 + b|$$

W.E-6: If the line $x \cos \alpha + y \sin \alpha = p$ touches the

hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$, then

$a^2 \cos^2 \alpha - b^2 \sin^2 \alpha$ is equal to

Sol: The given line is $x \cos \alpha + y \sin \alpha = p$

$$\Rightarrow y = -x \cot \alpha + p \operatorname{cosec} \alpha$$

On comparing with $y = mx + c$, we get

$$m = -\cot \alpha, c = p \operatorname{cosec} \alpha$$

Since, the given line touches the hyperbola

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1, \text{ then } c^2 = a^2 m^2 - b^2$$

$$\therefore p^2 \operatorname{cosec}^2 \alpha = a^2 \cot^2 \alpha - b^2$$

$$\Rightarrow a^2 \cos^2 \alpha - b^2 \sin^2 \alpha = p^2$$

W.E-7: The equations of the tangents to the hyperbola $x^2 - 4y^2 = 36$, which is perpendicular to the line $x - y + 4 = 0$ are

Sol: Let m be the slope of the tangent $x^2 - 4y^2 = 36$. Then, m (slope of the line perpendicular to

$$x - y + 4 = 0) \text{ is } -1. \text{ we have } \frac{x^2}{36} - \frac{y^2}{9} = 1$$

$$\therefore a^2 = 36 \text{ and } b^2 = 9$$

The equations of the tangents are

$$y = mx \pm \sqrt{a^2 m^2 - b^2}$$

$$\Rightarrow y = (-1)x \pm \sqrt{36 \times (-1)^2 - 9}$$

$$\Rightarrow y = -x \pm \sqrt{27} \Rightarrow x + y \pm 3\sqrt{3} = 0$$

W.E-8: The point of intersection of two tangents

to the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$, the product of whose slopes is c^2 ; lies on the curve

Sol: Product of the slopes of the tangents drawn from

$$p(x_1, y_1) \text{ to the hyperbola } \frac{x^2}{a^2} - \frac{y^2}{b^2} = 1,$$

$$\text{i.e., } m_1 m_2 = \frac{y_1^2 + b^2}{x_1^2 - a^2}$$

$$\therefore \frac{y_1^2 + b^2}{x_1^2 - a^2} = c^2 \Rightarrow (y_1^2 + b^2) = c^2 (x_1^2 - a^2)$$

$$\therefore \text{The locus of } p(x_1, y_1) \text{ is } y^2 + b^2 = c^2 (x^2 - a^2)$$

W.E-9: Let $P(a \sec \theta, b \tan \theta)$ and

$Q(a \sec \phi, b \tan \phi)$, where $\theta + \phi = \frac{\pi}{2}$, be two

points on the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$. If (h, k) is the point of intersection of the normals at P and Q , then k is equal to

Sol: Normal at $P(\theta)$ is

$$ax \cos \theta + by \cot \theta = a^2 + b^2 \quad \dots(i)$$

Normal at $Q(\theta)$ or $Q\left(\frac{\pi}{2} - \theta\right)$ is

$$ax \sin \theta + by \tan \theta = a^2 + b^2 \quad \dots(ii)$$

Both the normals passing through the point (h, k) .

$$\text{So } ah \cos \theta + bk \cot \theta = a^2 + b^2 \quad \dots(iii)$$

$$\text{and } ah \sin \theta + bk \tan \theta = a^2 + b^2 \quad \dots(iv)$$

Eliminating h from (iii) and (iv), we have

$$bk(\cot \theta \sin \theta - \tan \theta \cos \theta) = (a^2 + b^2)(\sin \theta - \cos \theta)$$

$$\Rightarrow k = -\left(\frac{a^2 + b^2}{b}\right)$$

W.E-10 : If the tangent and normal to $x^2 - y^2 = a^2$ at a point cut off intercepts a_1, a_2 and b_1, b_2 on the x-axis and y-axis respectively then $a_1 a_2 + b_1 b_2 =$

Sol. Let $p(x_1, y_1)$ is a point on $x^2 - y^2 = a^2$
Equation of the tangent at p is

$$xx_1 - yy_1 = a^2 \Rightarrow \frac{x}{\frac{a^2}{x_1}} + \frac{y}{-\frac{a^2}{y_1}} = 1$$

$$a_1 = \frac{a^2}{x_1}, b_1 = -\frac{a^2}{y_1},$$

Equation of the normal at p is

$$\frac{x}{x_1} + \frac{y}{y_1} = 2 \Rightarrow \frac{x}{2x_1} + \frac{y}{2y_1} = 1;$$

$$a_2 = 2x_1, b_2 = 2y_1$$

$$a_1 a_2 + b_1 b_2 = \frac{a^2}{x_1} \cdot (2x_1) + \left(-\frac{a^2}{y_1}\right)(2y_1) = 0$$

Asymptotes :

- The tangents at infinity to the hyperbola are called asymptotes of the hyperbola.
- The pair of equation of the asymptotes of the hyperbola $S = 0$ is $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 0$.
- The equation of a hyperbola and its pair of equation of asymptotes always differ by a constant.
- The asymptotes are always passing through the centre of the hyperbola.
- Asymptotes are equally inclined to the axes of hyperbola.
- The angle between the asymptotes of the hyperbola $S = 0$ is $2 \sec^{-1}(e)$ or $2 \tan^{-1}\left(\frac{b}{a}\right)$.
- The product of the perpendiculars from any point on the hyperbola to its asymptotes is $\frac{a^2 b^2}{a^2 + b^2}$.
- Any straight line parallel to an asymptotes of a hyperbola intersects the hyperbola in only one point.
- Asymptotes of a hyperbola and its conjugate hyperbola are same.

- If S, S' and A are the equations of a hyperbola, its conjugate hyperbola and its pair of asymptotes respectively then $S + S' = 2A$

➤ Rectangular Hyperbola $x^2 - y^2 = a^2$:

The hyperbola in which the lengths of the transverse and conjugate axes are equal is called an equilateral or rectangular hyperbola ($a = b$)

- (i) Eccentricity, $e = \sqrt{2}$
- (ii) The equation of the asymptotes are $y = \pm x$
- (iii) The angle between the asymptotes is 90° .

➤ Rectangular hyperbola $xy = c^2$:

If we rotate the axes through an angle of 45° in clock wise direction $x^2 - y^2 = a^2$ will become

$$xy = c^2, \left(c^2 = \frac{a^2}{2}\right). \quad \text{(i) centre } (0, 0)$$

$$\text{(ii) vertices } (\pm c, \pm c) \quad \text{(iii) Foci } (\pm\sqrt{2}c, \pm\sqrt{2}c)$$

$$\text{(iv) } e = \sqrt{2}$$

$$\text{(v) } LLR = 2\sqrt{2}c$$

$$\text{(vi) axes } y = x, y = -x$$

$$\text{(vii) Asymptotes } x = 0, y = 0$$

$$\text{(viii) Angle between the asymptotes} = 90^\circ$$

$$\text{(ix) Equation of tangent at } (x_1, y_1) \text{ is } xy_1 + x_1 y = 2c^2$$

$$\text{(x) Equation of normal at } (x_1, y_1) \text{ is } xx_1 - yy_1 = x_1^2 - y_1^2$$

$$\text{(xi) Parametric equations } x = ct, y = \frac{c}{t}, t \in R - \{0\}$$

$$\text{(xii) Equation of tangent at 't' is } x + yt^2 - 2ct = 0$$

$$\text{(xiii) Equation of normal at 't' is}$$

$$xt^3 - yt = c(t^4 - 1)$$

$$\text{(xiv) Normal at } t_1 \text{ meets the curve again at } t_2 \text{ then } t_1^3 t_2 = -1$$

$$\text{(xv) } P\left(ct_1, \frac{c}{t_1}\right), Q\left(ct_2, \frac{c}{t_2}\right), R\left(ct_3, \frac{c}{t_3}\right) \text{ are}$$

points on $xy = c^2$ then the ortho centre of ΔPQR

$$\text{is } S\left(\frac{-c}{t_1 t_2 t_3}, -ct_1 t_2 t_3\right)$$

Note : S also lies on $xy = c^2$

- (xiv) If a circle and a rectangular hyperbola $xy = c^2$ meet at four points

$$(x_i, y_i) = \left(ct_i, \frac{c}{t_i} \right), i = 1, 2, 3, 4 \text{ then } t_1 t_2 t_3 t_4 = 1$$

$$\Rightarrow x_1 x_2 x_3 x_4 = c^4, y_1 y_2 y_3 y_4 = c^4$$

Note:

$$P\left(ct_1, \frac{c}{t_1}\right), Q\left(ct_2, \frac{c}{t_2}\right), R\left(ct_3, \frac{c}{t_3}\right), S\left(ct_4, \frac{c}{t_4}\right)$$

are concyclic then the ortho centre of

$$\Delta PQR = \left(-ct_4, \frac{-c}{t_4} \right) = -S$$

Note: $x^2 - y^2 = a^2, xy = c^2$ cut each other orthogonally

- The hyperbola conjugate to $xy = c^2$ is $xy = -c^2$
- The equation of the hyperbola having $x = \alpha$ and $y = \beta$ as asymptotes can be taken as $(x - \alpha)(y - \beta) = k^2$
- Equation of the hyperbola with reference to two perpendicular lines $a_1x + b_1y + c_1 = 0$ and $b_1x - a_1y + c_2 = 0$ as axes of coordinates is

$$\frac{\left[\frac{a_1x + b_1y + c_1}{\sqrt{a_1^2 + b_1^2}} \right]^2}{a^2} - \frac{\left[\frac{b_1x - a_1y + c_2}{\sqrt{a_1^2 + b_1^2}} \right]^2}{b^2} = 1$$

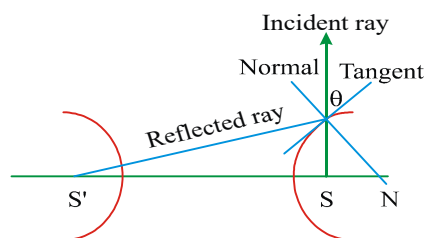
- The equation to the chord of contact of $P(x_1, y_1)$ with respect to the hyperbola $S = 0$ is $S_1 = 0$.
- The equation of the chord of the hyperbola $S = 0$ having $P(x_1, y_1)$ as its midpoint is $S_1 = S_{11}$
- The midpoint of the chord $lx + my + n = 0$ of the hyperbola $S = 0$ is

$$\left(-\frac{a^2 ln}{a^2 l^2 - b^2 m^2}, \frac{b^2 mn}{a^2 l^2 - b^2 m^2} \right)$$

- The equation of pair of tangents drawn from $P(x_1, y_1)$ to the hyperbola $S = 0$ is $S_1^2 = S.S_{11}$

➤ **Reflection property of a hyperbola :**

If an incoming light ray passing through one focus and strike convex side of the hyperbola then it will get reflected toward other focus S'



W.E-11 : $z = x + iy$ is a complex number. If the imaginary part of z^2 is 32, then locus of z is a hyperbola of eccentricity

Sol. $z = x + iy \Rightarrow z^2 = x^2 - y^2 + 2ixy$

$$\therefore \text{imaginary part of } z^2 = 32 \Rightarrow 2xy = 32$$

$$\Rightarrow xy = 16. \text{ It is a rectangular hyperbola } \Rightarrow e = \sqrt{2}$$

W.E-12 : The equation of a line passing through the centre of a rectangular hyperbola is $x - y - 1 = 0$. If one of the asymptotes is $3x - 4y - 6 = 0$, the equation of the other asymptote is

Sol. $\therefore$ Asymptotes are passing through centre. Point of intersection of $x - y - 1 = 0, 3x - 4y - 6 = 0$ is the centre i.e., $(-2, -3)$.

In a rectangular hyperbola asymptotes are perpendicular to each other.

$$\therefore \text{The equation of other asymptote is } 4x + 3y + k = 0$$

$$\therefore \text{By substituting centre } (-2, -3), k = 17$$

$$\therefore \text{Equation of other asymptote is } 4x + 3y + 17 = 0$$

W.E-13 : If a hyperbola passing through the origin has $3x - 4y - 1 = 0$ and $4x - 3y - 6 = 0$ as its asymptotes, then the equations of its transverse and conjugate axes are

Sol: The axes of a hyperbola are the bisectors of the pair of asymptotes.

Now, the bisectors of the given pair of asymptotes

$$\text{are } \frac{3x - 4y - 1}{5} = \pm \frac{4x - 3y - 6}{5}$$

$$\Rightarrow x + y - 5 = 0 \text{ and } x - y - 1 = 0$$

W.E-14 : If the tangents to the parabola $y^2 = 4ax$

intersect the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ at A and B , then the locus of point of intersection of tangents at A and B is

Sol. Let $P(x_1, y_1)$ be the point of intersection of tangents at A and B .

$\therefore AB$ is the chord of contact of P w.r. to the hyperbola. equation of AB is $S_1 = 0$

$$\Rightarrow \frac{xx_1}{a^2} - \frac{yy_1}{b^2} = 1 \Rightarrow y = \frac{(b^2 x_1)x}{a^2 y_1} - \frac{b^2}{y_1}$$

$\therefore$ it touches the parabola $y^2 = 4ax$

$$\Rightarrow C = \frac{a}{m} \Rightarrow -\frac{b^2}{y_1} = \frac{a^3 y_1}{b^2 x_1} \Rightarrow y_1^2 = -\frac{b^4}{a^3} x_1$$

$\therefore$ the locus of $P(x_1, y_1)$ is $y^2 = -\frac{b^4}{a^3} x$

W.E-15: The condition on ' a ' and ' b ' for which two distinct chords of the hyperbola

$\frac{x^2}{2a^2} - \frac{y^2}{2b^2} = 1$ passing through (a, b) are bisected by the line $x + y = b$ is

Sol. Let the line $x + y = b$ bisect the chord at $P(\alpha, b - \alpha)$. The equation of the chord whose midpoint is $P(\alpha, b - \alpha)$ is $S_1 = S_{11}$

$$\Rightarrow \frac{\alpha x}{2a^2} - \frac{(b - \alpha)y}{2b^2} = \frac{\alpha^2}{2a^2} - \frac{(b - \alpha)^2}{2b^2}$$

$\therefore$ it passes through (a, b)

$$\therefore \frac{\alpha}{2a} - \frac{(b - \alpha)}{2b} = \frac{\alpha^2}{2a^2} - \frac{(b - \alpha)^2}{2b^2}$$

$$\Rightarrow \alpha^2 \left(\frac{1}{a^2} - \frac{1}{b^2} \right) + \alpha \left(\frac{1}{b} - \frac{1}{a} \right) = 0$$

$$\Rightarrow \alpha = 0, \alpha = \frac{1}{\frac{1}{a} + \frac{1}{b}} = \frac{ab}{a + b}$$

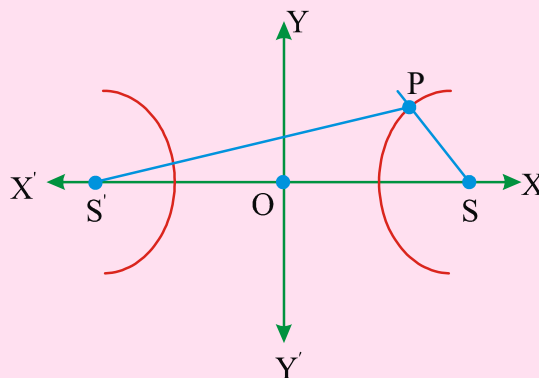
Hence, the required condition is $a + b \neq 0 \Rightarrow a \neq -b$

W.E-16 : If a ray of light incident along the line

$3x + (5 - 4\sqrt{2})y = 15$ gets reflected from the

hyperbola $\frac{x^2}{16} - \frac{y^2}{9} = 1$, then its reflected ray goes along the line

Sol.



The given hyperbola is $\frac{x^2}{16} - \frac{y^2}{9} = 1$

$$e = \sqrt{\frac{a^2 + b^2}{a^2}} = \frac{5}{4}, \text{ foci } = (\pm ae, 0) = (\pm 5, 0)$$

The ray is incident at $P(4\sqrt{2}, 3)$

The incident ray passes through $(5, 0)$, So the reflected ray must pass through $(-5, 0)$

$$\text{Equation of reflected ray is } y - 0 = \frac{3}{4\sqrt{2} + 5}(x + 5)$$

$$\Rightarrow 3x - (4\sqrt{2} + 5)y + 15 = 0$$

CONIC SECTION FORMULAE

Name of standard form of the conic	Parabola $S = y^2 - 4ax = 0$	Ellipse $S = \frac{x^2}{a^2} + \frac{y^2}{b^2} - 1 = 0$	Hyperbola $S = \frac{x^2}{a^2} - \frac{y^2}{b^2} - 1 = 0$
Centre (C)	—	C(0, 0)	C(0, 0)
Vertex (A)	(a, 0)	($\pm a$, 0)	($\pm a$, 0)
Axis	x-axis, $y = 0$	Major axis $y = 0$ Length = $2a$ Minor axis $x = 0$ Length = $2b$	Transverse axis $y = 0$ Length = $2a$ Conjugate axis $x = 0$ Length = $2b$
Eccentricity (e)	$e = 1$	$0 < e < 1, e = \sqrt{\frac{a^2 - b^2}{a^2}}$	$e > 1, e = \sqrt{\frac{a^2 + b^2}{a^2}}$
LLR (length of latus rectum)	$4a$	$\frac{2b^2}{a}$	$\frac{2b^2}{a}$
Focus (S)	(a, 0)	($\pm ae$, 0)	($\pm ae$, 0)
Directrix and foot of the directrix (Z)	$x = -a$, ($-a$, 0)	$x = \pm \frac{a}{e}, \left(\pm \frac{a}{e}, 0\right)$	$x = \pm \frac{a}{e}, \left(\pm \frac{a}{e}, 0\right)$
End point of latus rectum	(a, $\pm 2a$)	$\left(\pm ae, \pm \frac{b^2}{a}\right)$	$\left(\pm ae, \pm \frac{b^2}{a}\right)$
Equation of latus rectum	$x = a$	$x = \pm ae$	$x = \pm ae$
Focal distance of P(x_1 , y_1) (SP)	$x_1 + a$	$ a \pm ex_1 $	$ ex_1 \pm a $
Parametric equations	$x = at^2, y = 2at$	$x = a \cos \theta, y = b \sin \theta$	$x = a \sec \theta, y = b \tan \theta$
PSQ is a focal chord and 'l' is semi latus rectum then SP, l, SQ are in HP	$\frac{1}{SP} + \frac{1}{SQ} = \frac{2}{l}, l = 2a$	$\frac{1}{SP} + \frac{1}{SQ} = \frac{2}{l}, l = \frac{b^2}{a}$	$\frac{1}{SP} + \frac{1}{SQ} = \frac{2}{l}, l = \frac{b^2}{a}$
If $y = mx + C$ is a tangent then	$c = \frac{a}{m}$	$c^2 = a^2 m^2 + b^2$	$c^2 = a^2 m^2 - b^2$
Equation of tangent having slope m	$y = mx + \frac{a}{m}$	$y = mx \pm \sqrt{a^2 m^2 + b^2}$	$y = mx \pm \sqrt{a^2 m^2 - b^2}$
If $y = mx + c$ intersect the conic then	$cm < a$	$c^2 < a^2 m^2 + b^2$	$c^2 < a^2 m^2 - b^2$
Condition for $lx + my + n = 0$ is a tangent and point of contact	$ln = am^2$ $\left(\frac{n}{l}, \frac{2am}{l}\right)$	$a^2 l^2 + b^2 m^2 = n^2$ $\left(\frac{-a^2 l}{n}, \frac{-b^2 m}{n}\right)$	$a^2 l^2 - b^2 m^2 = n^2$ $\left(\frac{-a^2 l}{n}, \frac{b^2 m}{n}\right)$

Equation of normal having slope 'm'	$y = mx - 2am - am^3$	$y = mx \pm \frac{m(a^2 - b^2)}{\sqrt{a^2 + b^2m^2}}$	$y = mx \pm \frac{m(a^2 + b^2)}{\sqrt{a^2 - b^2m^2}}$
$lx + my + n = 0$ is a normal then	$al^3 + 2alm^2 + m^2n = 0$	$\frac{a^2}{l^2} + \frac{b^2}{m^2} = \frac{(a^2 - b^2)^2}{n^2}$	$\frac{a^2}{l^2} - \frac{b^2}{m^2} = \frac{(a^2 + b^2)^2}{n^2}$
Locus of point of intersection of perpendicular tangents	$x + a = 0$ (directrix)	$x^2 + y^2 = a^2 + b^2$ (Director circle)	$x^2 + y^2 = a^2 - b^2$ (Director circle)
Locus of the feet of the perpendicular from foci to any tangent	$x = a$ (tangent at the vertex)	$x^2 + y^2 = a^2$ (auxiliary circle)	$x^2 + y^2 = a^2$ (auxiliary circle)
m_1, m_2 are the slopes of tangents drawn from an external point $P(x_1, y_1)$ to the conic and ' θ ' is the angle between the tangents	$m_1 + m_2 = \frac{y_1}{x_1}$ $m_1 m_2 = \frac{a}{x_1}$ $\tan \theta = \frac{\sqrt{s_{11}}}{ x_1 + a }$	$m_1 + m_2 = \frac{2x_1 y_1}{x_1^2 - a^2}$ $m_1 m_2 = \frac{y_1^2 - b^2}{x_1^2 - a^2}$ $\tan \theta = \left \frac{2ab\sqrt{s_{11}}}{D_{11}} \right $ where D is the director circle	$m_1 + m_2 = \frac{2x_1 y_1}{x_1^2 - a^2}$ $m_1 m_2 = \frac{y_1^2 + b^2}{x_1^2 - a^2}$ $\tan \theta = \left \frac{2ab\sqrt{-s_{11}}}{D_{11}} \right $ D is the director circle
Equation of chord joining two points and chord passing through the focus then	$2x - (t_1 + t_2)y + 2at_1 t_2 = 0$ $t_1 t_2 = -1$	$\frac{x}{a} \cos \left(\frac{\alpha + \beta}{2} \right) + \frac{y}{b} \sin \left(\frac{\alpha + \beta}{2} \right) = \cos \left(\frac{\alpha - \beta}{2} \right)$ and $\tan \frac{\alpha}{2} \tan \frac{\beta}{2} = \frac{e-1}{e+1},$ $e = \frac{\cos \frac{\alpha - \beta}{2}}{\cos \frac{\alpha + \beta}{2}}$	$m_1 + m_2 = \frac{2x_1 y_1}{x_1^2 - a^2}$ $m_1 m_2 = \frac{y_1^2 + b^2}{x_1^2 - a^2}$ $\tan \theta = \left \frac{2ab\sqrt{-s_{11}}}{D_{11}} \right $ D is the director circle
P is a point on the conic	$SP = x_1 + a$ Tangent and normal at P meets the axis in T, N then $SP = ST = SN$	$SP + S'P = 2a$ and normal at P bisects $\angle SPS'$ and Tangent at P bisects external angle of $\angle SPS'$	$ SP - S'P = 2a$ and tangent at P bisects $\angle SPS'$ and normal at P bisects external angle of $\angle SPS'$
Maximum number of normals from a given point to the conic	3	4	4
Equation of tangent	At 't' $yt = x + at^2$	At ' θ ' $\frac{x}{a} \cos \theta + \frac{y}{b} \sin \theta = 1$	At ' θ ' $\frac{x}{a} \sec \theta - \frac{y}{b} \tan \theta = 1$
Equation of normal	At 't' $y + xt = 2at + at^3$	At ' θ ' $\frac{ax}{\cos \theta} - \frac{by}{\sin \theta} = a^2 - b^2$	At ' θ ' $\frac{ax}{\sec \theta} + \frac{by}{\tan \theta} = a^2 + b^2$

Product of the perpendicular from foci to any tangent is	—	b^2	b^2
Length of chord made by $y = mx + c$ on the conic	$\frac{4}{m^2} \times \sqrt{(1+m^2)a(a-mc)}$	$\frac{2ab}{a^2m^2+b^2} \times \frac{1}{\sqrt{(1+m^2)(a^2m^2+b^2-c^2)}}$	$\frac{2ab\sqrt{(1+m^2)(a^2m^2-b^2-c^2)}}{a^2m^2-b^2}$
Normal at $P(x_1, y_1)$ on the curve	$y - y_1 = \frac{-y_1}{2a}(x - x_1)$	$\frac{a^2x}{x_1} - \frac{b^2y}{y_1} = a^2 - b^2$	$\frac{a^2x}{x_1} + \frac{b^2y}{y_1} = a^2 + b^2$
Normal at θ on the curve	$y + xt = 2at + at^3$	$\frac{ax}{\cos \theta} - \frac{by}{\sin \theta} = a^2 - b^2$	$\frac{ax}{\sec \theta} + \frac{by}{\tan \theta} = a^2 + b^2$
Position of a point $P(x_1, y_1)$ with respect to the curve $S = 0$ i) P lies inside the curve ii) P lies on the curve iii) P lies outside the curve	$S_{11} < 0$ $S_{11} = 0$ $S_{11} > 0$	$S_{11} < 0$ $S_{11} = 0$ $S_{11} > 0$	$S_{11} > 0$ $S_{11} = 0$ $S_{11} < 0$
Reflection property	Any light ray is sent along a line parallel to the axis then the reflected ray passes through the focus	Incident ray from focus 'S' after reflection by ellipse at a point 'P' passes through the other focus S'	Incident ray from focus 'S' after reflection by hyperbola at a point 'P' passes through the other focus S'

Conic Section Important points

- General Equation of conic is
 $S = ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$
- $\Delta = abc + 2fgh - af^2 - bg^2 - ch^2$
- $\Delta = 0, h^2 - ab > 0, f^2 - bc > 0, g^2 - ac > 0$ then $S = 0$ represents a pair of lines
- $\Delta \neq 0, h = 0, a = b \neq 0$, then $S = 0$ represents a circle
- $\Delta \neq 0, h^2 - ab = 0$, then $S = 0$ represents a parabola
- $\Delta \neq 0, h^2 - ab < 0$, then $S = 0$ represents an Ellipse
- $\Delta \neq 0, h^2 - ab > 0$, then $S = 0$ represents a hyperbola
- Equation of tangent at $P(x_1, y_1)$ is $S_1 = 0$ (i.e.)
 $axx_1 + h[xy_1 + x_1y] + byy_1 + g(x + x_1) + f(y + y_1) + c = 0$
- Equation of normal at (x_1, y_1) is
 $\frac{x - x_1}{ax_1 + hy_1 + g} = \frac{y - y_1}{hx_1 + by_1 + f}$
- Solve $\frac{\partial S}{\partial x} = 0$, and $\frac{\partial S}{\partial y} = 0$ we get the centre of the conic (except parabola) (i.e.) $\left(\frac{hf - bg}{ab - h^2}, \frac{gh - af}{ab - h^2}\right)$

- Focus $S(x_1, y_1)$, directrix $lx + my + n = 0$, eccentricity 'e' equation of the conic is
 $SP^2 = e^2 PM^2$ (i.e.) $(x - x_1)^2 + (y - y_1)^2 = e^2 \left| \frac{lx + my + n}{\sqrt{l^2 + m^2}} \right|^2$
Axis is $m(x - x_1) + l(y - y_1) = 0$
- $LLR = 2e$ (The perpendicular distance from focus to the directrix) $= 2e(SZ)$
- (i) Axis of parabola and tangent at the vertex of parabola are $lx + my + n = 0, ax + by + c = 0$ then the equation of the parabola is
 $\left(\frac{lx + my + n}{\sqrt{l^2 + m^2}} \right)^2 = LLR \left(\frac{ax + by + c}{\sqrt{a^2 + b^2}} \right)$
(ii) In an ellipse major axis and its length are $lx + my + n_1 = 0$, $2a$ and minor axis and its length are $mx - ly + n_2 = 0$, $2b$ then the equation of ellipse is
 $\frac{\left(\frac{mx - ly + n_2}{\sqrt{l^2 + m^2}} \right)^2}{a^2} + \frac{\left(\frac{lx + my + n_1}{\sqrt{l^2 + m^2}} \right)^2}{b^2} = 1$
(iii) In a hyperbola transverse axis and its length are $lx + my + n_1 = 0$, $2a$ and conjugate axis and its length are $mx - ly + n_2 = 0$, $2b$ then the equation of hyperbola is
 $\frac{\left(\frac{mx - ly + n_2}{\sqrt{l^2 + m^2}} \right)^2}{a^2} - \frac{\left(\frac{lx + my + n_1}{\sqrt{l^2 + m^2}} \right)^2}{b^2} = 1$
- Normal of a conic passing through a focus does not exist (except axis of the conic)
- The eccentricity of circle is zero and whose directrix is at infinity distance

C.U.Q

- A point moves in a plane so that its distances PA and PB from two fixed points A and B in the plane satisfy the relation $|PA - PB| = K (K \neq 0)$, then the locus of P is
 - a hyperbola (if $AB > K$)
 - a branch of the hyperbola
 - a parabola
 - an ellipse
- Two straight lines passing through the fixed points $(\pm a, 0)$ and having slopes whose product is $k > 0$, then the locus of point of intersection of the lines is
 - A circle
 - A parabola
 - An ellipse
 - A hyperbola
- No part of the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ lies between which of the following
 - $x = -2a$ and $x = 2a$
 - $x = -a$ and $x = a$
 - $y = -2b$ and $y = 2b$
 - $y = -b$ and $y = b$
- Which of the following equations (t being the parameter) can't represent a hyperbola?
 - $\frac{tx}{a} - \frac{y}{b} + t = 0, \frac{x}{a} + \frac{ty}{b} - 1 = 0$
 - $x = \frac{a}{2}\left(t + \frac{1}{t}\right), y = \frac{b}{2}\left(t - \frac{1}{t}\right)$
 - $x = e^t + e^{-t}, y = e^t - e^{-t}$
 - $x^2 = 2(\cos t + 3), y^2 = 2\left(\cos^2 \frac{t}{2} - 1\right)$
- PN is the ordinate of any point P on the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ and AA^1 is its transverse axis. If Q divides AP in the ratio $a^2 : b^2$ then NQ is
 - perpendicular to AA^1
 - parallel to AA^1
 - perpendicular to OP
 - parallel to OP.

- P is a point on the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$, N is the foot of the perpendicular from P on the transverse axis. The tangent to the hyperbola at P meets the transverse axis at T. If O is the centre of the hyperbola, then the value of $OT \times ON$ is
 - a^2
 - b^2
 - $a^2 - b^2$
 - $a^2 + b^2$
- From any point $(x, y) \neq (\pm a, \pm b)$ on the hyperbola $x^2 - y^2 = a^2 - b^2$, two tangents are drawn to the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$. Then they make angles α and β with X-axis such that
 - $\tan \alpha - \tan \beta = 1$
 - $\tan \alpha + \tan \beta = 1$
 - $\tan \alpha \tan \beta = 1$
 - $\tan \alpha \tan \beta = 2$
- The locus of the point of intersection of the perpendicular tangents to the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ is
 - $x^2 + y^2 = a^2 + b^2$
 - $x^2 + y^2 = a^2 - b^2$
 - $x^2 + y^2 = a^2$
 - $x^2 + y^2 = b^2$
- Radius of director circle of a rectangular hyperbola is
 - 5
 - 10
 - 1
 - 0
- If the line $lx + my = 1$ is a normal to the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$, then $\frac{a^2}{l^2} - \frac{b^2}{m^2}$ is equal to (EAM-2007)
 - $a^2 - b^2$
 - $a^2 + b^2$
 - $(a^2 + b^2)^2$
 - $(a^2 - b^2)^2$
- Normal at t_1 on $xy = c^2$ meets the curve again at t_2 then $t_1^3 t_2 =$
 - 1
 - 2
 - 1
 - 2
- If the circle $x^2 + y^2 = a^2$ intersects the hyperbola $xy = c^2$ in four points (x_i, y_i) , for $i = 1, 2, 3$ and 4, then $y_1 + y_2 + y_3 + y_4 =$ (EAM-2009)
 - 0
 - c
 - a
 - c^4

13. If the circle $x^2 + y^2 = a^2$ intersects the hyperbola $xy = c^2$ in four points

$P(x_1, y_1), Q(x_2, y_2), R(x_3, y_3)$ and

$S(x_4, y_4)$, then

- 1) $x_1 + x_2 + x_3 + x_4 = 1$
- 2) $y_1 + y_2 + y_3 + y_4 = 1$
- 3) $x_1 x_2 x_3 x_4 = c^2$ 4) $y_1 y_2 y_3 y_4 = c^4$

14. If a triangle is inscribed in a rectangular hyperbola, its orthocentre lies

- 1) inside the curve 2) outside the curve
- 3) on the curve 4) can't be determined

15. Given the base of a triangle and the ratio of the tangents of half the base angle. Then the vertex lies on

- 1) a parabola 2) an ellipse
- 3) a hyperbola 4) Circle

16. Two circles are given such that they neither intersect nor touch. Then the locus of centre of variable circle which touches both the circles externally.

- 1) circle 2) parabola
- 3) ellipse 4) hyperbola

C.U.Q - KEY

- | | | | | | |
|-------|-------|-------|-------|-------|-------|
| 1) 1 | 2) 4 | 3) 2 | 4) 1 | 5) 1 | 6) 1 |
| 7) 3 | 8) 2 | 9) 4 | 10) 3 | 11) 3 | 12) 1 |
| 13) 4 | 14) 3 | 15) 3 | 16) 4 | | |

C.U.Q - HINTS

1. $|SP - S^1P| = 2a$ represents a hyperbola.
2. $\frac{y-0}{x-a} \times \frac{y-0}{x+a} = k$
3. Hyperbola does not lie between the tangents at vertices
4. Eliminating t .
5. $A = (a, 0), A^1 = (-a, 0)$ let $P = (a \sec \theta, b \tan \theta)$
 $N = (a \sec \theta, 0)$
 $AQ : QP = a^2 : b^2$ find Q and apply
 (slope of NQ).(slope of A^1P) = -1

6. $P(x_1, y_1)$ be a point on the hyperbola .

Then the coordinates of N are $(x_1, 0)$. The equation of the tangent at (x_1, y_1) is

$$\frac{xx_1}{a^2} - \frac{yy_1}{b^2} = 1 \text{ This meets x-axis at } T\left(\frac{a^2}{x_1}, 0\right)$$

$$\therefore OT.ON = \frac{a^2}{x_1} x_1 = a^2$$

7. $(y_1 - mx_1)^2 = a^2 m^2 + b^2$

$$m_1 m_2 = \frac{y_1^2 - b^2}{x_1^2 - a^2}, \quad \tan \alpha . \tan \beta = 1$$

8. director circle

9. Director circle of $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ is

$$x^2 + y^2 = a^2 - b^2. \text{ In a rectangular hyperbola } a = b$$

$$10. \frac{a^2}{l^2} - \frac{b^2}{m^2} = \frac{(a^2 + b^2)^2}{n^2}$$

11. Normal at t_1

$$xt_1^3 - yt_1 = c(t_1^4 - 1)$$

$$\text{Passes through } \left(ct_2, \frac{c}{t_2}\right) \quad ct_2 t_1^3 - \frac{ct_1}{t_2} = c(t_1^4 - 1)$$

$$t_2^3 t_1^3 - t_1 = t_1^4 t_2 - t_2 \quad t_1^3 t_2 (t_2 - t_1) = t_1 - t_2$$

$$t_1^3 t_2 = -1$$

$$12. x = \frac{c^2}{y}, x^2 + y^2 = a^2 \Rightarrow \frac{c^2}{y^2} + y^2 = a^2$$

$$y^4 - a^2 y^2 + c^4 = 0$$

$$\text{Sum of the roots } y_1 + y_2 + y_3 + y_4 = 0$$

$$13. x = \frac{c^2}{y}, x^2 + y^2 = a^2 \Rightarrow \frac{c^2}{y^2} + y^2 = a^2$$

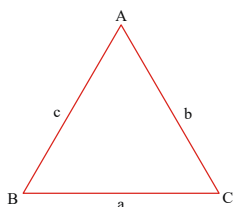
$$y^4 - a^2 y^2 + c^4 = 0$$

$$\text{Now product of roots } y_1 y_2 y_3 y_4 = c^4$$

14. Rectangular hyperbola $xy = c^2$ orthocentre of the triangle whose vertices t_1, t_2, t_3 is

$$O\left(\frac{-c}{t_1 t_2 t_3}, -c t_1 t_2 t_3\right).$$

15.



$$\text{Also } \frac{\tan \frac{B}{2}}{\tan \frac{C}{2}} = k (\text{constant})$$

$$\Rightarrow \frac{\sqrt{\frac{s(s-b)}{(s-a)(s-c)}}}{\sqrt{\frac{s(s-c)}{(s-a)(s-b)}}} = k \Rightarrow \frac{s-b}{s-c} = k$$

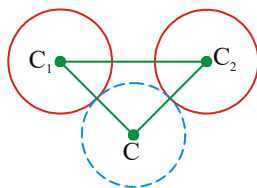
$$\Rightarrow \frac{(s-b) - (s-c)}{(s-b) + (s-c)} = \frac{k-1}{k+1}$$

$$\Rightarrow c-b = a \frac{k-1}{k+1} = \text{constant}$$

$$\Rightarrow AB - AC = \text{constant}$$

So, locus of A is a hyperbola.

16.



In the figure circles with solid line have centre C_1 and C_2 and radii r_1 and r_2 . Let the circle which touches given two circles as explained in the question which has centre C and radius r.

Now, $CC_2 = r + r_2$ and $CC_1 = r_1 + r$

Hence, $CC_1 - CC_2 = r_1 - r_2 (= \text{constant})$

Then locus of C is hyperbola whose foci are C_1 and C_2 .



LEVEL-I (C.W)

1. The locus of the point of intersection of the straight lines $\frac{x}{a} + \frac{y}{b} = \lambda$ and $\frac{x}{a} - \frac{y}{b} = \frac{1}{\lambda}$ (λ is variable) is

- 1) a circle 2) a parabola
3) an Ellipse 4) a hyperbola

2. The equation $2x^2 + 3y^2 - 8x - 18y + 35 = k$ represents

- 1) no locus if $k > 0$ 2) an ellipse if $k < 0$
3) a point if $k = 0$ 4) a hyperbola if $k > 0$

3. For the hyperbola $\frac{x^2}{\cos^2 \alpha} - \frac{y^2}{\sin^2 \alpha} = 1$

which of the following remains constant when α varies? (AIEEE-2007)

- 1) abscissae of foci 2) eccentricity
3) directrix 4) abscissae of vertices

CENTRE, FOCI, LENGTH OF LATUS RECTUM AND AXES

4. The eccentricity of a hyperbola is $\frac{\sqrt{5}}{2}$,

then its length of latusrectum is

- 1) $\frac{1}{4}(2a)$ 2) $\frac{1}{2}(2a)$ 3) $\frac{2}{3}(2a)$ 4) $\frac{1}{3}(2a)$

5. In a hyperbola $e = \frac{9}{4}$ and the distance

between the directrices is 3. Then the length of transverse axis is

- 1) $27/2$ 2) $27/8$ 3) $27/4$ 4) $17/4$

6. Equation of the transverse axis of the

$$\text{hyperbola } \frac{(y-2)^2}{9} - \frac{(x+3)^2}{16} = 1 \text{ is}$$

- 1) $x+3=0$ 2) $x-5=0$
3) $x-7=0$ 4) $x-9=0$

7. The length of the latusrectum of the hyperbola $9x^2 - 16y^2 + 72x - 32y - 16 = 0$ is

- 1) $\frac{9}{2}$ 2) $\frac{32}{3}$ 3) $\frac{11}{5}$ 4) $\frac{21}{5}$

ECCENTRICITY

8. In a hyperbola the distance between the foci is three times the distance between the directrices then its eccentricity is
 1) $\frac{5}{2}$ 2) $\frac{3}{2}$ 3) $\frac{5}{4}$ 4) $\sqrt{3}$
9. If the latusrectum of a hyperbola subtends a right angle at its centre then its $e =$
 1) $\frac{\sqrt{3}+1}{2}$ 2) $\frac{\sqrt{7}+1}{2}$ 3) $\frac{\sqrt{5}+1}{2}$ 4) $\frac{\sqrt{5}}{2}$
10. If the length of latusrectum of a hyperbola $\frac{x^2}{k} - \frac{y^2}{25} = -1$ is $\frac{22}{5}$ then its e is
 1) $\frac{7}{3}$ 2) $\frac{7}{5}$ 3) $\frac{6}{5}$ 4) $\frac{7}{2}$
11. The eccentricity of the hyperbola $25x^2 - y^2 + 200x - 375 = 0$ is
 1) $2\sqrt{5}$ 2) $2\sqrt{6}$ 3) $\sqrt{22}$ 4) $\sqrt{26}$
12. If $\sec \theta$ is the eccentricity of a hyperbola then the eccentricity of the conjugate hyperbola is
 1) $\tan \theta$ 2) $\cot \theta$ 3) $\cos \theta$ 4) $\operatorname{cosec} \theta$

EQUATION OF HYPERBOLA

13. Equation of the hyperbola with $e = \sqrt{2}$ and having the distance between the foci 1 is
 1) $x^2 - y^2 = \frac{1}{4}$ 2) $x^2 - y^2 = 5$
 3) $x^2 - y^2 = \frac{1}{6}$ 4) $x^2 - y^2 = \frac{1}{8}$
14. Equation of the hyperbola with foci $(0, \pm 5)$ and $C = \frac{5}{3}$ is
 1) $\frac{x^2}{9} - \frac{y^2}{16} = 1$ 2) $\frac{x^2}{16} - \frac{y^2}{9} = -1$
 3) $\frac{x^2}{16} - \frac{y^2}{9} = 1$ 4) $\frac{x^2}{12} - \frac{y^2}{13} = 1$

15. Equation of the locus of all points such that the difference of its distances from $(-3, -7)$, $(-3, 3)$ is 8 is

1) $\frac{(x+3)^2}{16} - \frac{(y+2)^2}{9} = 1$
 2) $\frac{(x+3)^2}{9} - \frac{(y+2)^2}{16} = -1$
 3) $\frac{(x+3)^2}{9} - \frac{(y+2)^2}{19} = 1$
 4) $\frac{(x+3)^2}{7} - \frac{(y+2)^2}{19} = -1$

16. Equation of the hyperbola with vertices $(\pm 2, 0)$ foci $(\pm 3, 0)$

1) $5x^2 - 4y^2 = 20$ 2) $5x^2 + 4y^2 = 20$
 3) $4x^2 - 5y^2 = 20$ 4) $5x^2 - 4y^2 = -20$

17. A hyperbola passes through a focus of the

ellipse $\frac{x^2}{169} + \frac{y^2}{25} = 1$. Its transverse and conjugate axes coincide respectively with the major and minor axes of the ellipse. The product of eccentricities is 1. Then the equation of the hyperbola is (EAM-2014)

1) $\frac{x^2}{144} - \frac{y^2}{25} = 1$ 2) $\frac{x^2}{25} - \frac{y^2}{9} = 1$
 3) $\frac{x^2}{144} - \frac{y^2}{9} = 1$ 4) $\frac{x^2}{169} - \frac{y^2}{25} = 1$

EQUATION OF DIRECTRIX AND LATUSRECTUM

18. The equation of the latusrecta of the

hyperbola $\frac{(x-4)^2}{16} - \frac{(y-3)^2}{20} = 1$ are

1) $x = 1 \pm 5$ 2) $x = 4 \pm 6$
 3) $y = 2 \pm 6$ 4) $y = 3 \pm 5$

19. Equation of one of the latusrectum of the hyperbola

$(10x - 5)^2 + (10y - 2)^2 = 9(3x + 4y - 7)^2$ is

1) $30x + 40y - 23 = 0$
 2) $40x + 30y - 23 = 0$
 3) $30x + 40y + 23 = 0$
 4) $40x + 30y + 23 = 0$

20. The foci of the hyperbola are $S(-3, -2)$, $S^1(5, 6)$. If its $e=2$ then the equation of its directrix corresponding to focus S is
 1) $x + y - 3 = 0$ 2) $x + y - 5 = 0$
 3) $x + y - 7 = 0$ 4) $x + y - 1 = 0$
21. The centre, one vertex, one focus of a hyperbola are $(1, -1), (5, -1), (6, -1)$ its directrices are
 1) $5x - 21 = 0, 5x + 11 = 0$.
 2) $5x + 4 = 0, x + 1 = 0$
 3) $5x - 7 = 0, 5x - 19 = 0$
 4) $5x - 17 = 0, 5x - 6 = 0$

TANGENTS AND NORMALS

22. The condition that the line $\frac{x}{p} + \frac{y}{q} = 1$ to

be tangent to $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ is

- 1) $\frac{a^2}{p^2} + \frac{b^2}{q^2} = 1$ 2) $\frac{a^2}{p^2} - \frac{b^2}{q^2} = 1$
 3) $\frac{a^2}{q^2} - \frac{b^2}{p^2} = 1$ 4) $\frac{a^2}{q^2} + \frac{b^2}{p^2} = 1$

23. A tangent to the hyperbola $3x^2 - 4y^2 = 12$ makes equal intercepts on the axes. Then the area of the triangle formed by the tangent with coordinate axes is
 1) 2 2) $1/4$ 3) 4 4) $1/2$
24. If $4x - 3y = \sqrt{3}$ is a tangent to $4x^2 - 9y^2 = 1$ then the eccentric angle of the point of contact is
 1) $\frac{\pi}{4}$ 2) $\frac{\pi}{6}$ 3) $\frac{\pi}{3}$ 4) $\frac{\pi}{2}$
25. The slopes of the common tangents to the parabola $y^2 = 24x$ and the hyperbola $5x^2 - y^2 = 5$ are
 1) ± 3 2) ± 2 3) ± 6 4) ± 5

26. Number of tangents drawn from $(-2, -1)$ to $2x^2 - 3y^2 = 6$ are

- 1) 3 2) 0 3) 1 4) 2

27. Total number of tangents of the hyperbola $\frac{x^2}{9} - \frac{y^2}{4} = 1$, that are perpendicular to the line $5x + 2y - 3 = 0$ is/are

- 1) 0 2) 1 3) 2 4) 4

28. If m_1 and m_2 are the slopes of tangents to the hyperbola $\frac{x^2}{25} - \frac{y^2}{16} = 1$ which pass through the

point $(6, 2)$ then $\frac{11m_1m_2}{10} =$

- 1) 1 2) 2 3) 3 4) 4

29. Equation of one of the tangents passing through $(2, 8)$ to the hyperbola $5x^2 - y^2 = 5$ is (EAM-2012)

- 1) $3x + y - 14 = 0$ 2) $3x - y + 2 = 0$
 3) $x + y + 3 = 0$ 4) $x - y + 6 = 0$

RECTANGULAR AND CONJUGATE HYPERBOLA

30. If $5x^2 + \lambda y^2 = 20$ represents a rectangular hyperbola, then λ is equal to
 1) 5 2) 4 3) -5 4) -4

31. The equation to the conjugate hyperbola of $xy + 3x - 4y + 13 = 0$ is

- 1) $(x - 4)(y + 3) = 0$
 2) $xy + 3x - 4y - 13 = 0$
 3) $(x - 4)(y + 3) = 25$
 4) $(x - 4)(y + 3) = -25$

32. The equation of the director circle of the hyperbola $\frac{x^2}{16} - \frac{y^2}{25} = 1$ is

- 1) $x^2 + y^2 = 9$ 2) $x^2 + y^2 = 16$
 3) $x^2 + y^2 = 25$ 4) does not exist

33. The product of the perpendiculars from the foci on any tangent to the hyperbola

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1 \text{ is}$$

- 1) b^2 2) a^2 3) $-b^2$ 4) $2b^2$

34. The length of the transverse axis of the rectangular hyperbola $xy = 18$ is

- 1) 6 2) 12 3) 18 4) 9

ASYMPTOTES

35. The asymptotes of the hyperbola are $3x = \pm 5y$ and its vertices are $(\pm 5, 0)$ then the length of latusrectum of the hyperbola is

- 1) $9/5$ 2) $18/5$ 3) $50/3$ 4) $25/3$

36. If θ is the angle between the asymptotes of the hyperbola

$$x^2 + 2xy - 3y^2 + x + 7y + 9 = 0 \text{ then}$$

$$\tan \theta =$$

- 1) $2/3$ 2) $1/5$ 3) 2 4) $4/5$

37. The coordinates of a point on the hyperbola,

$$\frac{x^2}{24} - \frac{y^2}{18} = 1, \text{ which is nearest to the line}$$

$$3x + 2y + 1 = 0 \text{ are}$$

- 1) $(6, 3)$ 2) $(-6, -3)$ 3) $(6, -3)$ 4) $(-6, 3)$

38. The transverse axis of a hyperbola is of length $2a$ and a vertex divides the segment of the axis between the centre and the corresponding focus in the ratio 2:1. The equation of the hyperbola is

- 1) $5x^2 - 4y^2 = 5a^2$ 2) $5x^2 - 4y^2 = 4a^2$
3) $4x^2 - 5y^2 = 5a^2$ 4) $4x^2 - 5y^2 = 4a^2$

LEVEL-I (C.W.) - KEY

- 1) 4 2) 3 3) 1 4) 1 5) 3 6) 1
7) 1 8) 4 9) 3 10) 3 11) 4 12) 4
13) 4 14) 2 15) 2 16) 1 17) 1 18) 2
19) 1 20) 4 21) 1 22) 2 23) 4 24) 2
25) 1 26) 4 27) 1 28) 2 29) 2 30) 3
31) 3 32) 4 33) 1 34) 2 35) 2 36) 3
37) 3 38) 1

LEVEL-I (C.W.) - HINTS

$$1. \left(\frac{x}{a} + \frac{y}{b} \right) \left(\frac{x}{a} - \frac{y}{b} \right) = \lambda \cdot \frac{1}{\lambda}$$

$$2. 2(x-2)^2 + 3(y-3)^2 = k$$

For $k = 0$, we get $2(x-2)^2 + 3(y-3)^2 = 0$
which represents the point $(2, 3)$

$$3. e = \sqrt{\frac{a^2 + b^2}{a^2}} = \sec \alpha; \text{ foci} = (\pm ae, 0) = (\pm 1, 0)$$

$$4. e = \frac{\sqrt{5}}{2}; \frac{2b^2}{a} = \frac{2a^2(e^2 - 1)}{a} = 2a \left(\frac{5}{4} - 1 \right) = \frac{1}{4}(2a)$$

$$5. e = \frac{9}{4}, \frac{2a}{e} = 3, \text{ length of transverse axis} = 2a$$

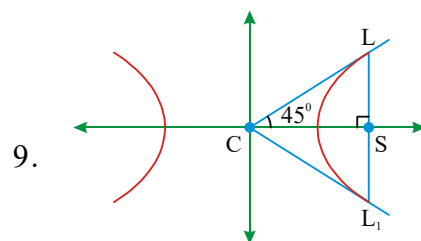
$$6. x - h = 0$$

$$7. 9x^2 - 16y^2 + 72x - 32y - 16 = 0$$

$$\Rightarrow 9(x-4)^2 - 16(y+1)^2 = 144 \Rightarrow \frac{(x-4)^2}{16} - \frac{(y+1)^2}{9} = 1$$

$$\text{Length of latusrectum } \frac{2b^2}{a} = \frac{9}{2}$$

$$8. 2ae = 3 \left(2 \cdot \frac{a}{e} \right) \Rightarrow e^2 = 3$$



$$\text{from triangle CSL, } \tan 45^\circ = \frac{b^2/a}{ae}$$

$$a^2e = b^2 \Rightarrow e^2 - e - 1 = 0 \quad e = \frac{\sqrt{5} + 1}{2}$$

$$10. \frac{2a^2}{b} = \frac{22}{5} \text{ and } \frac{2k}{5} = \frac{22}{5} \Rightarrow k = 11; e = \sqrt{\frac{a^2 + b^2}{b^2}}$$

$$11. 25(x^2 + 8x + 16 - 16) - y^2 - 375 = 0$$

$$\frac{(x+4)^2}{31} - \frac{y^2}{775} = 1$$

$$a^2 = 31, b^2 = 775, e = \sqrt{\frac{a^2 + b^2}{a^2}}$$

$$12. \frac{1}{e_1^2} + \frac{1}{e_2^2} = 1$$

$$13. 2ae = 1 \text{ and } e = \sqrt{2} \Rightarrow a = \frac{1}{2\sqrt{2}}$$

$$14. S(0, \pm be) = (0, \pm 5), e = \frac{5}{3}, a^2 = b^2(e^2 - 1)$$

$$15. x\text{-coordinates of foci are equal}$$

$$|SP - S^1P| = 2b = 8 \Rightarrow b = 4, SS^1 = 2be$$

$$a^2 = b^2e^2 - b^2, \text{ centre is midpoint of } S \text{ \& } S'$$

$$16. y\text{-coordinates of foci and vertices are zero}$$

$$a = 2, ae = 3 \text{ using } b^2 = a^2e^2 - a^2$$

$$17. e_1 = \sqrt{\frac{169-25}{169}} = \frac{12}{13}; e_2 = \sqrt{\frac{144+25}{144}} = \frac{13}{12} \quad e_1e_2 = 1$$

$$18. \text{Centre } C(h, k) = (4, 3), e = \sqrt{\frac{a^2+b^2}{a^2}} = \frac{3}{2}$$

$$\text{Equation of latusrecta } x = h \pm ae \Rightarrow x = 4 \pm 6$$

$$19. 100 \left[\left(x - \frac{1}{2} \right)^2 + \left(y - \frac{1}{5} \right)^2 \right] = 9(3x + 4y - 7)^2 S \left(\frac{1}{2}, \frac{1}{5} \right),$$

$$\text{directrix is } 3x + 4y - 7 = 0, \text{ latus rectum}$$

$$3x + 4y + k = 0 \text{ passing through } \left(\frac{1}{2}, \frac{1}{5} \right), k = \frac{-23}{10}$$

$$20. CS : SZ = e^2 : 1 - e^2 = 4 : -3 \Rightarrow Z = (0, 1) \text{ lies on the directrix}$$

$$21. y\text{-coordinates of centre, vertex, focus are equal}$$

$$a = 4, ae = 5 \Rightarrow e = \frac{5}{4} \quad x = 1 \pm \frac{16}{5}$$

$$22. a^2\ell^2 - b^2m^2 = n^2$$

$$23. x + y = a, m = -1, y = mx \pm \sqrt{a^2m^2 - b^2}$$

$$24. P(\theta) = \left(\frac{\sec \theta}{2}, \frac{\tan \theta}{3} \right); S_1 = 0 \text{ and } m = \frac{4}{3}$$

$$25. \text{Equation of tangent to parabola}$$

$$y^2 = 24x (a = 6) \text{ is } y = mx + \frac{6}{m} \Rightarrow m^2x - my + 6 = 0$$

$$\text{It is also tangent to hyperbola } \frac{x^2}{1} - \frac{y^2}{5} = 1$$

$$a^2\ell^2 - b^2m^2 = n^2$$

$$26. S_{11} < 0$$

$$27. \text{perpendicular slope } m = \frac{2}{5}, a^2m^2 - b^2 < 0$$

$$28. m_1m_2 = \frac{y_1^2 + b^2}{x_1^2 - a^2} = \frac{20}{11}$$

$$29. \text{Equation of the tangent is } y = mx \pm \sqrt{a^2m^2 - b^2}$$

$$\text{Passing through } (2, 8) \Rightarrow 8 = 2m \pm \sqrt{m^2 - 5}$$

$$\Rightarrow (8 - 2m)^2 = m^2 - 5$$

$$30. ax^2 + by^2 = k \text{ represents a rectangular hyperbola then } a = -b.$$

$$31. \text{Given eq. of the hyperbola } (x-4)(y+3) = -25$$

$$\text{Req. eq. of the hyperbola } (x-4)(y+3) = 25$$

$$32. x^2 + y^2 = a^2 - b^2 \text{ and } a^2 - b^2 < 0 \dots$$

$$33. \text{The product of the perpendiculars from the foci on any tangent to the hyperbola } S = 0 \text{ is } b^2.$$

$$34. \text{Vertices } A(C, C), A'(-C, -C), \text{ length of transverse axis} = AA'$$

$$35. \text{Asymptotes are } \frac{x}{5} = \pm \frac{y}{3}$$

$$\text{Equation of hyperbola } \frac{x^2}{25} - \frac{y^2}{9} = 1; LLR = 2\frac{b^2}{a}.$$

$$36. \tan \theta = \frac{2\sqrt{h^2 - ab}}{a + b}.$$

$$37. P(x_1, y_1) \text{ be the point on the hyperbola nearest to the given line. } \therefore \frac{x_1^2}{24} - \frac{y_1^2}{18} = 1 \dots (i)$$

$$\text{Tangent at P is parallel to the given line}$$

$$\left(\frac{dy}{dx} \right) p(x_1, y_1) = \frac{-3}{2}; x = -2y \dots (ii)$$

$$\text{Solving (i) \& (ii)}$$

$$38. CA : AS = 2 : 1 \Rightarrow CA = 2AS$$

$$\Rightarrow CA = 2(CS - CA) \Rightarrow a = 2(ae - a)$$

$$\Rightarrow e = \frac{3}{2}; \quad b^2 = a^2(e^2 - 1)$$



LEVEL-I (H.W)

CENTRE, FOCI, LENGTH OF LATUS RECTUM AND AXES

$$1. \text{The length of latusrectum of the hyperbola } 25y^2 - 16x^2 = 400 \text{ is}$$

$$1) \frac{8}{3} \quad 2) \frac{9}{2} \quad 3) \frac{11}{3} \quad 4) \frac{25}{2}$$

$$2. \text{The distance between the foci of the hyperbola } \frac{x^2}{16} - \frac{y^2}{9} = 1 \text{ is}$$

$$1) 2\sqrt{7} \quad 2) 10 \quad 3) 8 \quad 4) 6$$

3. In the hyperbola the length of conjugate axis is 5 and the distance between the foci is 13 then the length of transverse axis is
1) 10 2) 14 3) 16 4) 12
4. The distance between the foci of the hyperbola $x^2 - 3y^2 - 4x - 6y - 11 = 0$ is (EAM-2008)
1) 4 2) 6 3) 8 4) 10

ECCENTRICITY

5. If the eccentricity of a conic is $\frac{3}{\sqrt{5}}$ then the conic is
1) an ellipse 2) a hyperbola
3) a parabola 4) a circle
6. e and e^1 are the eccentricities of the hyperbolas $16x^2 - 9y^2 = 144$ and $9x^2 - 16y^2 = -144$ then $e - e^1 =$
1) 2 2) 1 3) 0 4) $\frac{3}{2}$
7. In a hyperbola the latusrectum equals to semi transverse axis, its eccentricity is
1) $\frac{3}{2}$ 2) $\frac{4}{3}$ 3) $\frac{5}{2}$ 4) $\sqrt{\frac{3}{2}}$
8. The ratio between the transverse axis and conjugate axes of a hyperbola is 5:2 then its $e =$
1) $\frac{\sqrt{31}}{5}$ 2) $\frac{\sqrt{29}}{5}$ 3) $\frac{\sqrt{33}}{5}$ 4) $\frac{\sqrt{29}}{4}$
9. The eccentricity of a hyperbola is $\frac{5}{3}$ then the eccentricity of the conjugate hyperbola is
1) $\frac{5}{2}$ 2) $\frac{7}{2}$ 3) $\frac{7}{3}$ 4) $\frac{5}{4}$
10. If the eccentricity of a hyperbola is $\sqrt{3}$; then the eccentricity of its conjugate hyperbola is (EAM-2006)
1) $\sqrt{2}$ 2) $\sqrt{3}$ 3) $\frac{\sqrt{3}}{2}$ 4) $2\sqrt{3}$

EQUATION OF HYPERBOLA

11. Equation of the hyperbola with vertex (4,0) and focus (6,0) is
1) $\frac{x^2}{16} - \frac{y^2}{20} = 1$ 2) $\frac{x^2}{20} - \frac{y^2}{16} = 1$
3) $\frac{x^2}{16} - \frac{y^2}{36} = 1$ 4) $\frac{x^2}{36} - \frac{y^2}{16} = 1$

12. Equation of the hyperbola with centre (0,0) distance between the foci 18 and distance between directrices 8 is

1) $\frac{x^2}{36} - \frac{y^2}{45} = 1$ 2) $\frac{x^2}{45} - \frac{y^2}{36} = 1$
3) $\frac{x^2}{64} - \frac{y^2}{32} = 1$ 4) $\frac{x^2}{32} - \frac{y^2}{65} = 1$

13. Equation of the hyperbola with focus (-3,4) directrix $3x - 4y + 5 = 0$ and $e = \frac{5}{2}$ is

1) $5x^2 - 24xy + 12y^2 + 6x - 8y - 75 = 0$
2) $5x^2 - 24xy + 12y^2 - 8x - 6y - 25 = 0$
3) $5x^2 - 24xy + 12y^2 - 12x + 8y - 55 = 0$
4) $5x^2 - 24xy + 12y^2 - 7x - 12y - 65 = 0$

14. Equation of the hyperbola with foci $(\pm 5, 0)$ and the transverse axis of the length 8 is

1) $9x^2 - 16y^2 = 144$ 2) $9x^2 + 16y^2 = 144$
3) $3x^2 + 4y^2 = 12$ 4) $3x^2 - 4y^2 = 12$

15. The equation of the hyperbola whose foci are $(-2, 0)$ and $(2, 0)$ and eccentricity is 2 is given by (EAM-2011)

1) $-3x^2 + y^2 = 3$ 2) $x^2 - 3y^2 = 3$
3) $3x^2 - y^2 = 3$ 4) $-x^2 + 3y^2 = 3$

EQUATION OF DIRECTRIX AND LATUS RECTUM

16. The equations of the latusrecta of the hyperbola $3y^2 - 4x^2 = 12$ are

1) $x = \pm\sqrt{11}$ 2) $y = \pm\sqrt{3}$
3) $y = \pm\sqrt{7}$ 4) $x = \pm\sqrt{5}$

17. Equations of the directrices of the hyperbola

$9x^2 - 16y^2 + 72x - 32y - 16 = 0$ are

1) $x - 1 = 0, x + 9 = 0$
2) $x + 4 = 0, x + 1 = 0$
3) $x = 0, x = 8$
4) $5x + 36 = 0, 5x + 4 = 0$

TANGENTS AND NORMALS

18. Equation of the tangent to the conic

$x^2 - y^2 - 8x + 2y + 11 = 0$ at (2,1) is

1) $x + 2 = 0$ 2) $2x + y - 5 = 0$
3) $x + y - 3 = 0$ 4) $x - 2 = 0$

19. Equation of the normal to the hyperbola

$$3x^2 - y^2 = 3 \text{ at } (2, -3) \text{ is}$$

- 1) $x - 2y - 8 = 0$ 2) $3x - 2y - 12 = 0$
3) $x + 2y + 4 = 0$ 4) $3x + 2y - 14 = 0$

20. If the line $y = x + c$ is a tangent to the

$$\text{hyperbola } \frac{x^2}{25} - \frac{y^2}{9} = 1 \text{ then } c \text{ is}$$

- 1) ± 4 2) ± 5 3) ± 3 4) ± 6

21. Sum of the slopes of the two tangents

$$\text{drawn from } (-2, -1) \text{ to } 2x^2 - 3y^2 = 6$$

- 1) 4 2) $9/2$ 3) $7/2$ 4) 7

22. Equation of the tangent to the hyperbola

$$3x^2 - y^2 = 3 \text{ parallel to the line } y = 2x + 4 \text{ is}$$

- 1) $y = 2x + 3$ 2) $y = 2x - 3$ 3) $y = 2x + 1$ 4) $y = 2x + 5$

23. If the tangent at the point $(2 \sec \theta, 3 \tan \theta)$

$$\text{of the hyperbola } \frac{x^2}{4} - \frac{y^2}{9} = 1 \text{ is parallel}$$

$$\text{to } 3x - y + 4 = 0, \text{ then the value of } \theta \text{ is}$$

- 1) 45° 2) 60° 3) 30° 4) 75°

24. The value of m , for which the line

$$y = mx + \frac{25\sqrt{3}}{3} \text{ is a normal to the conic}$$

$$\frac{x^2}{16} - \frac{y^2}{9} = 1 \text{ is}$$

- 1) $-\frac{2}{\sqrt{3}}$ 2) $\sqrt{3}$ 3) $-\frac{\sqrt{3}}{2}$ 4) $\frac{1}{\sqrt{3}}$

RECTANGULAR AND CONJUGATE HYPERBOLA

25. If e and e_1 are the eccentricities of the hyperbola and its conjugate then

$$e_1^2 - e^2 =$$

- 1) $e^2 e_1^2 - 2e^2$ 2) $e^2 e_1^2 + 2e^2$
3) $2e^2 - e^2 e_1^2$ 4) $2e^2 + e_1^2$

26. If e and e_1 are the eccentricities of

$$xy = c^2, \quad x^2 - y^2 = a^2 \text{ then } e^4 + e_1^4 =$$

- 1) 9 2) 64 3) 8 4) 81

27. Foci of curve $xy = 4$ are

- 1) $(\sqrt{2}, -\sqrt{2}), (-\sqrt{2}, -\sqrt{2})$
2) $(2\sqrt{2}, 2\sqrt{2}), (-2\sqrt{2}, -2\sqrt{2})$
3) $(-2\sqrt{2}, 2\sqrt{2}), (2\sqrt{2}, -2\sqrt{2})$
4) $(-\sqrt{3}, \sqrt{3}), (\sqrt{3}, -\sqrt{3})$

28. The auxiliary circle of the hyperbola

$$xy = 9 \text{ is}$$

- 1) $x^2 + y^2 = 9$ 2) $x^2 + y^2 = 6$
3) $x^2 + y^2 = 18$ 4) $x^2 + y^2 = 36$

AUXILIARY AND DIRECTOR CIRCLES

29. If SK is the perpendicular from focus S

$$\text{of the hyperbola } \frac{x^2}{12} - \frac{y^2}{6} = 1 \text{ on any tangent to it then K lies on}$$

- 1) $x^2 + y^2 = 12$ 2) $x^2 + y^2 = 18$
3) $x^2 + y^2 = 6$ 4) $x^2 + y^2 = 9$

30. Equation of the director circle of the

$$\text{hyperbola } \frac{x^2}{25} - \frac{y^2}{9} = 1 \text{ is}$$

- (1) $x^2 + y^2 = 18$ (2) $x^2 + y^2 = 9$
(3) $x^2 + y^2 = 25$ (4) $x^2 + y^2 = 16$

ASYMPTOTES

31. If the angle between the asymptotes of a hyperbola is 30° then eccentricity =

- 1) $\sqrt{6} + \sqrt{2}$ 2) $\sqrt{6} - \sqrt{2}$
3) $\sqrt{5} - \sqrt{3}$ 4) $\sqrt{5} + \sqrt{3}$

32. If $3x - 2y + 7 = 0$ and $2x + 3y + 4 = 0$ are the asymptotes of a hyperbola then its eccentricity is

- 1) $\frac{3}{2}$ 2) $\sqrt{2}$ 3) 2 4) can't be determined

33. The angle between the asymptotes of the hyperbola $x^2 - 3y^2 = 3$ is (EAM-2011)

- 1) $\frac{\pi}{6}$ 2) $\frac{\pi}{4}$ 3) $\frac{\pi}{3}$ 4) $\frac{\pi}{2}$

34. The asymptotes of the hyperbola

$$xy = hx + ky \text{ are}$$

- 1) $x = k, y = h$ 2) $x = h, y = k$
3) $x = h, y = h$ 4) $x = k, y = k$

LEVEL-I (H.W.) - KEY

- 1) 4 2) 2 3) 4 4) 3 5) 2 6) 3 7) 4
8) 2 9) 4 10) 3 11) 1 12) 1 13) 1 14) 1
15) 3 16) 3 17) 4 18) 4 19) 1 20) 1 21) 1
22) 3 23) 3 24) 1 25) 1 26) 3 27) 2 28) 3
29) 1 30) 4 31) 2 32) 2 33) 3 34) 1

LEVEL-I (H.W.) - HINTS

- $25y^2 - 16x^2 = 400 \Rightarrow \frac{x^2}{25} - \frac{y^2}{16} = -1$
Length of latusrectum $= \frac{2a^2}{b} = \frac{25}{2}$
- $d = 2ae, a = 4, e = \frac{5}{4}$
- $2b = 5, 2ae = 13$
- $x^2 - 3y^2 - 4x - 6y - 11 = 0$
 $\Rightarrow \frac{(x-2)^2}{12} - \frac{(y+1)^2}{4} = 1$ distance between foci $= 2ae$
- $e = \frac{3}{\sqrt{5}} > 1$
- $e = \sqrt{\frac{a^2 + b^2}{a^2}}, e' = \sqrt{\frac{a^2 + b^2}{b^2}}$
- $\frac{2b^2}{a} = a \Rightarrow 2a^2(e^2 - 1) = a^2$
- $a : b = 5 : 2 \Rightarrow \frac{a^2}{b^2} = \frac{25}{4} \Rightarrow \frac{a^2}{a^2(e^2 - 1)} = \frac{25}{4}$
- $e = \frac{5}{3}, \frac{1}{e^2} + \frac{1}{e_1^2} = 1$
- $e_1 = \sqrt{3}; \frac{1}{e_1^2} + \frac{1}{e_2^2} = 1$
- $(a, 0) = (4, 0), (ae, 0) = (6, 0); b^2 = a^2e^2 - a^2$
- $2ae = 18, \frac{2a}{e} = 8 \Rightarrow ae \cdot \frac{a}{e} = 9 \times 4 \Rightarrow a = 6$
- $S(-3, 4), L \equiv 3x - 4y + 5 = 0, e = \frac{5}{2}, p(x_1, y_1)$
 $SP^2 = e^2 PM^2$
- y - coordinates of foci are zero
 $ae = 5$ and $2a = 8, b^2 = a^2e^2 - a^2$
- y - coordinate of foci are zero
 $SS' = 2ae = 4 \Rightarrow a = 1; b^2 = a^2(e^2 - 1) = 3$
- $3y^2 - 4x^2 = 12 \Rightarrow \frac{x^2}{3} - \frac{y^2}{4} = -1; e = \sqrt{\frac{a^2 + b^2}{b^2}} = \frac{\sqrt{7}}{2}$
Equation of latusrecta $y = \pm be \Rightarrow y = \pm \sqrt{7}$
- $9(x^2 + 8x + 16 - 16) - 16(y^2 + 2y + 1 - 1) - 16 = 0$
 $\Rightarrow \frac{(x+4)^2}{16} - \frac{(y+1)^2}{9} = 1$

Equation of directrices $x = h \pm \frac{a}{e}$

- $S_1 = 0$
- $\frac{a^2x}{x_1} + \frac{b^2y}{y_1} = a^2 + b^2$
- $c = \pm \sqrt{a^2m^2 - b^2}$
- $m_1 + m_2 = \frac{2x_1y_1}{x_1^2 - a^2}$
- $y = mx \pm \sqrt{a^2m^2 - b^2}$ where $m = 2$
- $S_1 = 0$ and $m = 3$
- $mx - y + \frac{25\sqrt{3}}{3} = 0; \frac{a^2}{l^2} - \frac{b^2}{m^2} = \frac{(a^2 + b^2)^2}{n^2}$
- $e_1^2 + e^2 = e^2e_1^2 \Rightarrow e_1^2 - e^2 = e^2 \cdot e_1^2 - 2e^2$
- $e = \sqrt{2}, e_1 = \sqrt{2}$
- Foci of $xy = c^2$ $(\pm\sqrt{2}c, \pm\sqrt{2}c)$
- Auxiliary circle of $xy = c^2$ is $x^2 + y^2 = 2c^2$
- K lies on auxiliary circle, $x^2 + y^2 = a^2$
- $x^2 + y^2 = a^2 - b^2$
- Angle between asymptotes
 $\theta = 2\sec^{-1}e \Rightarrow e = \sec \frac{\theta}{2}$
- Given asymptotes are perpendicular $\Rightarrow$ Hyperbola is a rectangular hyperbola; $e = \sqrt{2}$
- $\theta = 2\tan^{-1}\left(\frac{b}{a}\right)$
- Equation of hyperbola $xy = hx + ky$
 $\Rightarrow (x-k)(y-h) = hk$

Pair of equation of asymptotes $(x-k)(y-h) = 0$



LEVEL-II (C.W)

- The equation $\left| \sqrt{(x-2)^2 + (y-1)^2} - \sqrt{(x+2)^2 + y^2} \right| = c$ will represent a hyperbola if
1) $c \in (0, 6)$ 2) $c \in (0, 5)$
3) $c \in (0, \sqrt{17})$ 4) $c \in (0, \sqrt{19})$
- The product of focal distances of the point (4, 3) on the hyperbola $x^2 - y^2 = 7$ is
1) 25 2) 12 3) 9 4) 16

3. The eccentricity of the conic

$$4(2y-x-3)^2 - 9(2x+y-1)^2 = 80 \text{ is}$$

- 1) $\frac{3}{13}$ 2) $\frac{\sqrt{13}}{3}$ 3) $\sqrt{13}$ 4) 3

ECCENTRICITY

4. PQ is a double ordinate of the hyperbola

$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ such that OPQ is equilateral triangle, O being the centre of the hyperbola, then find range of the eccentricity e of the hyperbola.

- 1) $1 < e < 2$ 2) $1 < e < \sqrt{2}$
3) $e \geq \frac{2}{\sqrt{3}}$ 4) $e < 4$

5. The differential equation $\frac{dx}{dy} = \frac{3y}{2x}$ represent a family of hyperbolas (except when it represents a pair of lines) with eccentricity

- 1) $\sqrt{\frac{3}{5}}$ 2) $\sqrt{2}$ 3) $\sqrt{\frac{2}{5}}$ 4) $\sqrt{\frac{5}{2}}$

6. The two curves $\frac{x^2}{16} - \frac{y^2}{9} = 1$ and $\frac{x^2}{a^2} + \frac{y^2}{9} = 1$ touch each other then the ratio of their eccentricities is

- 1) 2:1 2) $\sqrt{2}:1$ 3) 5:7 4) $5:\sqrt{7}$

CENTRE, FOCI

7. The foci of the hyperbola

$$9x^2 - 18x - 16y^2 - 64y + 89 = 0 \text{ are}$$

- 1) $(1, 3), (1, -7)$ 2) $(1, 5), (1, -9)$
3) $(1, 1), (1, -5)$ 4) $(1, 5), (1, -11)$

8. S and S^1 are the foci and A is one vertex of $x^2 - 2y^2 = 1$ then $SA \cdot S^1A =$

- 1) 2 2) 1 3) $\frac{1}{2}$ 4) $\frac{1}{4}$

9. The equation of hyperbola with coordinate axes as principal axes, distance of one of its

vertices from foci is 9 and 1 units is $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$

and $a+b=$

1. 5 2. 7 3. 9 4. 6

10. If P is a point on the rectangular hyperbola $x^2 - y^2 = a^2$, " C " is its centre and S, S^1 are foci then $SP \cdot S^1P =$

- 1) 2 2) $(CP)^2$ 3) $(CS)^2$ 4) $(SS^1)^2$

TANGENTS AND NORMALS

11. Equation of the common tangent to $y^2 = 8x$ and $3x^2 - y^2 = 3$ is

- 1) $2x - y + 1 = 0$ 2) $x + 2y - 5 = 0$
3) $2x + y - 6 = 0$ 4) $2x - y + 7 = 0$

12. If the line $ax + by + c = 0$ is a normal to the hyperbola $xy = 1$, then

- 1) $a > 0, b < 0$ 2) $a > 0, b > 0$
3) $a < 0, b < 0$ 4) $a = 0, b = 0$

13. The equations of the common tangents to

the two hyperbolas $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ and

$\frac{y^2}{a^2} - \frac{x^2}{b^2} = 1$ are :

- (1) $y = \pm x \pm \sqrt{b^2 - a^2}$ (2) $y = \pm x \pm \sqrt{a^2 - b^2}$
(3) $y = \pm x \pm (a^2 - b^2)$ (4) $y = \pm x \pm \sqrt{a^2 + b^2}$

14. The locus of a point $P(\alpha, \beta)$ moving under the condition that the line $y = \alpha x + \beta$ is a tangent to the hyperbola

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1 \text{ is}$$

- 1) a parabola 2) an ellipse 3) a hyperbola 4) a circle

AUXILIARY AND DIRECTOR CIRCLES

15. If the tangent at the point (h, k) on the

hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ meets the auxiliary circle of the hyperbola in two points whose

ordinates y_1, y_2 then $\frac{1}{y_1} + \frac{1}{y_2} =$

- 1) $\frac{1}{k}$ 2) $\frac{1}{k^2}$ 3) $\frac{4}{k^2}$ 4) $\frac{2}{k}$

ASYMPTOTES

16. Area of the triangle formed by any

tangent to the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ with its asymptotes is

- 1) ab 2) $2ab$ 3) $2a^2b^2$ 4) a^2b^2

17. The area (in square units) of the equilateral triangle formed by the tangent at $(\sqrt{3}, 0)$ to the hyperbola $x^2 - 3y^2 = 3$ with the pair of asymptotes of the hyperbola is (EAM-2012)

- 1) $\sqrt{2}$ 2) $\sqrt{3}$ 3) $\frac{1}{\sqrt{3}}$ 4) $2\sqrt{3}$

CHORD OF CONTACT AND MIDPOINT OF CHORD

18. If $x=9$ is the chord of contact of the hyperbola $x^2 - y^2 = 9$, then the equation of the tangent at one of the points of contact is (EAM-2013)

- 1) $x + \sqrt{3}y + 2 = 0$ 2) $3x - 2\sqrt{2}y - 3 = 0$
3) $3x - \sqrt{2}y + 6 = 0$ 4) $x - \sqrt{3}y + 2 = 0$

19. The locus of middle points of chords of hyperbola $3x^2 - 2y^2 + 4x - 6y = 0$ parallel to $y = 2x$ is

- 1) $3x - 4y = 4$ 2) $3y - 4x + 4 = 0$
3) $4x - 3y = 3$ 4) $3x - 4y = 2$

20. Equation of the chord of the hyperbola $25x^2 - 16y^2 = 400$, which is bisected at the point $(6, 2)$, is

- 1) $16x - 75y = 418$ 2) $75x - 16y = 418$
3) $25x - 4y = 400$ 4) $4y + 25x = 100$

21. The equation of the chord joining two points (x_1, y_1) and (x_2, y_2) on the rectangular hyperbola $xy = c^2$ is

- 1) $\frac{x}{x_1 + x_2} + \frac{y}{y_1 + y_2} = 1$ 2) $\frac{x}{x_1 - x_2} + \frac{y}{y_1 - y_2} = 1$
3) $\frac{x}{y_1 + y_2} + \frac{y}{x_1 + x_2} = 1$ 4) $\frac{x}{y_1 - y_2} + \frac{y}{x_1 - x_2} = 1$

22. The mid-point of the chord $4x - 3y = 5$ of the hyperbola $2x^2 - 3y^2 = 12$ is (EAM-2009)

- 1) $(0, -\frac{5}{3})$ 2) $(2, 1)$ 3) $(\frac{5}{4}, 0)$ 4) $(\frac{11}{4}, 2)$

23. If the chord of contact of tangents from a point P to the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ subtends a right angle at the centre, then the locus of P is

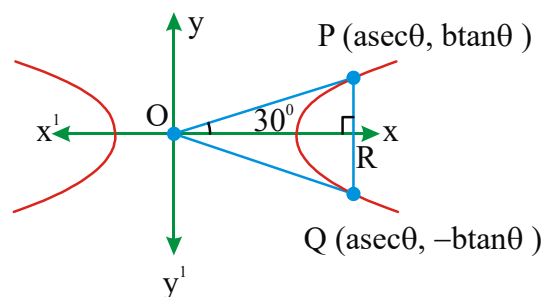
- 1) a parabola 2) an ellipse
3) a hyperbola 4) a circle

LEVEL-II (C.W.) - KEY

- 1) 3 2) 1 3) 2 4) 3 5) 4 6) 4
7) 1 8) 3 9) 2 10) 2 11) 1 12) 1
13) 2 14) 3 15) 4 16) 1 17) 2 18) 2
19) 1 20) 2 21) 1 22) 2 23) 2

LEVEL-II (C.W.) - HINTS

1. $S(2,1), S^1(-2,0)$ and $C < SS^1 \Rightarrow C < \sqrt{17}$
2. $SP \cdot S^1P = (CP)^2$ where C is centre.
3. $\frac{\left(\frac{2y-x-3}{\sqrt{5}}\right)^2}{4} - \frac{\left(\frac{2x+y-1}{\sqrt{5}}\right)^2}{16/9} = 1 \therefore e = \sqrt{\frac{a^2+b^2}{a^2}}$
4. Let double ordinate PQ be such that $P = (a \sec \theta, b \tan \theta)$, and $Q(a \sec \theta, -b \tan \theta)$ and O is centre $(0,0)$.



From $\triangle OPR \Rightarrow \tan 30^\circ = \frac{b \tan \theta}{a \sec \theta} \Rightarrow 3 \frac{b^2}{a^2} = \cos^2 \theta$

$$\Rightarrow 3(e^2 - 1) = \cos^2 \theta \geq 1 \Rightarrow e \geq \frac{2}{\sqrt{3}}$$

5. $\frac{dx}{dy} = \frac{3y}{2x} \Rightarrow \int 2x dx = \int 3y dy$

$$\Rightarrow x^2 = \frac{3y^2}{2} + c \Rightarrow \frac{x^2}{1} - \frac{y^2}{2/3} = c$$

If c is positive, then $e = \sqrt{1 + \frac{2}{3}} = \sqrt{\frac{5}{3}}$

If c is negative, then $e = \sqrt{1 + \frac{3}{2}} = \sqrt{\frac{5}{2}}$

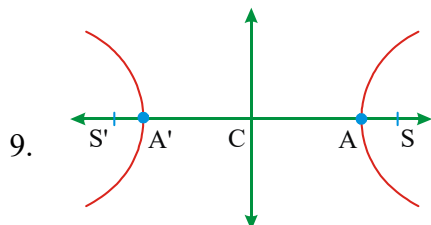
6. Slopes of the tangents at $P(x_1, y_1)$ to the two curves are equal $\Rightarrow \frac{9x_1}{16y_1} = \frac{9x_1}{a^2 y_1} \Rightarrow a^2 = 16$

7. $9(x^2 - 2x + 1 - 1) - 16(y^2 + 4y + 4 - 4) + 89 = 0$

$$\Rightarrow \frac{(x-1)^2}{16} - \frac{(y+2)^2}{9} = -1, \text{ foci} = S(h, k \pm be)$$

8. $S(ae, 0), S^1(-ae, 0), A(a, 0)$

$$SA \cdot S^1A = a^2e^2 - a^2 = b^2 = \frac{1}{2}$$



$$AS^1 = 9$$

$$AS = 1$$

$$\Rightarrow AA^1 + A^1S^1 = 9$$

$$\Rightarrow CS - CA = 1$$

$$\Rightarrow 2a + 1 = 9$$

$$\Rightarrow ae - a = 1$$

$$\Rightarrow a = 4$$

$$\Rightarrow e = \frac{5}{4}$$

$$b^2 = a^2(e^2 - 1) = 9$$

10. $SPS^1P = (ePM).(ePM^1)$

11. $y = mx + \frac{a}{m}, a = 2$ and $c^2 = a^2m^2 - b^2$

12. Slope of tangent $\frac{dy}{dx} = \frac{-1}{x^2}$

Slope of normal $= x^2 > 0 \quad \therefore -\frac{a}{b} > 0$

13. Any tangent to the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ is

$$y = mx \pm \sqrt{a^2m^2 - b^2}$$

or $y = mx + c$ where $c = \pm \sqrt{a^2m^2 - b^2} \rightarrow (1)$

This will touch the hyperbola $\frac{y^2}{a^2} - \frac{x^2}{b^2} = 1$, if the

equation $\frac{(mx+c)^2}{a^2} - \frac{x^2}{b^2} = 1$ has equal

roots or $x^2(b^2m^2 - a^2) + 2b^2mcx + (c^2 - a^2)b^2 = 0$ has equal roots.

$$\Rightarrow 4b^2m^2c^2 = 4(b^2m^2 - a^2)(c^2 - a^2)b^2$$

$$\Rightarrow c^2 = a^2 - b^2m^2$$

from (1) $\therefore a^2m^2 - b^2 = a^2 - b^2m^2$

$$\Rightarrow m^2(a^2 + b^2) = a^2 + b^2 \Rightarrow m = \pm 1$$

Hence, the equation of the common tangents are

$$y = \pm x \pm \sqrt{a^2 - b^2}$$

14. $c^2 = a^2m^2 - b^2$

15. Equation of the tangent at (h, k) is

$$\frac{hx}{a^2} - \frac{ky}{b^2} = 1 \Rightarrow x = \frac{a^2}{h} \left(1 + \frac{yk}{b^2} \right)$$

Equation of auxiliary circle $x^2 + y^2 = a^2 \frac{1}{y_1} + \frac{1}{y_2} = \frac{y_1 + y_2}{y_1 y_2}$

16. Equation of tangent at $P(\theta)$ is

$$\frac{x \sec \theta}{a} - \frac{y \tan \theta}{b} = 1$$

Pair of equation of

asymptotes $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 0$. Area of $\Delta l e = \frac{n^2 \sqrt{h^2 - ab}}{|am^2 - 2hlm + bl^2|}$

17. Area of the triangle $= ab$

18. Solve $x = 9$ and $x^2 - y^2 = 9$ equation of tangent at point of contact is $S_1 = 0$

19. Equation of the chord, $P(x_1, y_1)$ as its midpoint is $S_1 = S_{11}$. Slope of the chord $= 2$

20. Equation of the chord $S_1 = S_{11}$

21. Midpoint of the chord $= \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$

Equation of the chord is $S_1 = S_{11}$

22. $\left(\frac{-a^2 \ln}{a^2 l^2 - b^2 m^2}, \frac{-b^2 mn}{a^2 l^2 - b^2 m^2} \right)$

23. $S_1 = 0$ and homogenisation $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1, a + b = 0$



LEVEL-II (H.W)

1. If the equation

$$(10x - 5)^2 + (10y - 4)^2 = \lambda^2 (3x + 4y - 1)^2$$

represents a hyperbola then:

(1) $-2 < \lambda < 2$ (2) $\lambda > 2$

(3) $\lambda < -2$ or $\lambda > 2$ (4) $0 < \lambda < 2$

2. The equation

$$2x^2 + 5xy + 2y^2 - 11x - 7y - 4 = 0$$

represents

1) a pair of lines

2) Parabola

3) Ellipse

4) Hyperbola

ECCENTRICITY

3. The eccentricity of the hyperbola $4x^2 - 9y^2 = 2ax + b^2$ is (E-1999)
 1) $\frac{a}{b}$ 2) $\frac{\sqrt{b}}{a}$ 3) $\frac{\sqrt{13}}{2}$ 4) $\frac{\sqrt{13}}{3}$
4. If the eccentricity of the hyperbola $x^2 - y^2 \operatorname{cosec}^2 \alpha = 25$ is $\sqrt{5}$ times the eccentricity of the ellipse $x^2 \operatorname{cosec}^2 \alpha + y^2 = 5$, then α is equal to :
 1) $\tan^{-1} \sqrt{2}$ 2) $\sin^{-1} \sqrt{\frac{3}{4}}$ 3) $\tan^{-1} \sqrt{\frac{2}{5}}$ 4) $\sin^{-1} \sqrt{\frac{2}{5}}$
5. Let e_1, e_2 are eccentricities of $\frac{x^2}{16} - \frac{y^2}{9} = 1$ and $\frac{x^2}{16} + \frac{y^2}{b^2} = 1$ and $e_1 e_2 = 1$ the foci of the ellipse
 1) $(\pm \frac{9}{5}, 0)$ 2) $(\pm \frac{4}{5}, 0)$ 3) $(\pm \frac{15}{3}, 0)$ 4) $(\pm \frac{16}{5}, 0)$

CENTRE, FOCI

6. The distance between the foci of the hyperbola $9x^2 - 16y^2 + 18x + 32y - 151 = 0$ is
 1) 2 2) 8 3) 10 4) 6
7. The equation of the transverse axis of the hyperbola $(x-3)^2 + (y+1)^2 = (4x+3y)^2$ is
 1. $x+3y=0$ 2. $4x+3y=9$
 3. $3x-4y=13$ 4. $4x+3y=0$
8. If the foci of the ellipse $\frac{x^2}{25} + \frac{y^2}{16} = 1$ and the hyperbola $\frac{x^2}{4} - \frac{y^2}{b^2} = 1$ coincide, then $b^2 =$ (EAM-2013)
 1) 4 2) 5 3) 8 4) 9

TANGENTS AND NORMALS

9. Equation of the tangent to the hyperbola $4x^2 - 3y^2 = 24$ which make an angle 60° with x - axis is
 1) $y = x\sqrt{3} \pm 12$ 2) $y = x\sqrt{3} \pm \sqrt{10}$
 3) $y = x\sqrt{3} \pm 9$ 4) $y = x\sqrt{3} \pm 5$
10. Number of normals to a hyperbola which passing through a focus are (except transverse axis)
 1) 1 2) 2 3) ∞ 4) 0

AUXILIARY AND DIRECTOR CIRCLES

11. A circle cuts perpendicular lines so that each of intercept is of given length (but not equal) then eccentricity of the locus of the centre of the circle
 1. $\frac{1}{\sqrt{2}}$ 2. 1 3. $\sqrt{2}$ 4. 2
12. If the vertex of a hyperbola bisects the distance between its centre and the corresponding focus, then ratio of square of its conjugate axis to the square of its transverse axis is
 1. 2 2. 4 3. 6 4. 3

ASYMPTOTES

13. The equation of the hyperbola which passes through the point (2, 3) and has the asymptotes $4x+3y-7=0$ and $x-2y-1=0$ is (EAM-2010)
 1) $4x^2 + 5xy - 6y^2 - 11x + 11y + 50 = 0$
 2) $4x^2 + 5xy - 6y^2 - 11x + 11y - 43 = 0$
 3) $4x^2 - 5xy - 6y^2 - 11x + 11y + 57 = 0$
 4) $x^2 - 5xy - y^2 - 11x + 11y - 43 = 0$

LEVEL-II (H.W.) - KEY

- 1) 3 2) 4 3) 4 4) 1 5) 4 6) 3 7) 3
 8) 2 9) 2 10) 4 11) 3 12) 4 13) 3

LEVEL-II (H.W.) - HINTS

1.
$$\frac{\sqrt{\left(x-\frac{1}{2}\right)^2 + \left(y-\frac{2}{5}\right)^2}}{\left|\frac{3x+4y-1}{5}\right|} = \left|\frac{\lambda}{2}\right|,$$

 Here $\left|\frac{\lambda}{2}\right| > 1 \Rightarrow |\lambda| > 2 \Rightarrow \lambda < -2 \text{ or } \lambda > 2$
2. $\Delta \neq 0, h^2 - ab > 0.$
3. Equation of hyperbola will be $\frac{\left(x-\frac{a}{4}\right)^2}{\frac{4b^2+a^2}{16}} - \frac{y^2}{\frac{4b^2+a^2}{36}} = 1$
4. Eccentricity of $\frac{x^2}{25} - \frac{y^2}{25\sin^2 \alpha} = 1$ is $\sqrt{1+\sin^2 \alpha}$
 Eccentricity of $\frac{x^2}{5\sin^2 \alpha} + \frac{y^2}{5} = 1$ is $\sqrt{1-\sin^2 \alpha}$
 $\Rightarrow \sin^2 \alpha = \frac{2}{3} \Rightarrow \alpha = \sin^{-1} \sqrt{\frac{2}{3}} = \tan^{-1} \sqrt{2}$

$$5. e_1 = \sqrt{\frac{a^2 + b^2}{a^2}} = \frac{5}{4}; e_2 = \sqrt{\frac{a^2 - b^2}{a^2}} = \frac{4}{5} \text{ foci } (\pm ae, 0)$$

$$6. 9(x^2 + 2x + 1 - 1) - 16(y^2 - 2y + 1 - 1) - 151 = 0$$

$$\Rightarrow \frac{(x+1)^2}{16} - \frac{(y-1)^2}{9} = 1, e = \sqrt{\frac{a^2 + b^2}{a^2}}$$

distance between foci = $2ae$

7. Focus $(3, -1)$, directrix $4x + 3y = 0$ transverse axis is perpendicular to directrix and passing through focus

$$8. e_1 = \sqrt{\frac{a^2 - b^2}{a^2}} = \frac{3}{5}, \text{ foci } = (\pm ae, 0) = (\pm 3, 0)$$

$$e_2 = \sqrt{\frac{a^2 + b^2}{a^2}} = \sqrt{\frac{4 + b^2}{4}}; \therefore 2\sqrt{\frac{4 + b^2}{4}} = 3$$

$$9. m = \sqrt{3}; y = mx \pm \sqrt{a^2 m^2 - b^2}$$

10. Number of normals of a hyperbola which passing through a focus is 1 i.e., transverse axis

$$11. C(x_1, y_1), 2\sqrt{g^2 - c_1} = 2\sqrt{f^2 - c_2}$$

$$x_1^2 - c_1 = y_1^2 - c_2; x^2 - y^2 = k$$

$$12. \text{ Let the hyperbola be } \frac{x^2}{a^2} - \frac{y^2}{b^2} = 1 \text{ then}$$

$$2a = ae, \text{ i.e., } e = 2 \therefore \frac{b^2}{a^2} = e^2 - 1 = 3 \therefore \frac{(2b)^2}{(2a)^2} = 3$$

13. Equation of hyperbola and pair of equation of asymptotes differ by constant



LEVEL-III

1. The chord of the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$,

whose equation is $x \cos \alpha + y \sin \alpha = P$, subtends a right angle at its centre and it always touches a circle of radius

$$1) \frac{ab}{\sqrt{b^2 - a^2}} \quad 2) \frac{a^2 b^2}{\sqrt{b^2 - a^2}} \quad 3) \frac{a^2 b^2}{\sqrt{b - a}} \quad 4) \frac{ab}{\sqrt{b - a}}$$

2. If values of m for which the line $y = mx + 2\sqrt{5}$ touches the hyperbola $16x^2 - 9y^2 = 144$ are roots of the equation $x^2 - (a+b)x - 4 = 0$, then value of $(a+b)$ is equal to

$$1. 2 \quad 2. 4 \quad 3. \text{ zero} \quad 4. 3$$

3. The equations of the transverse and conjugate axes of a hyperbola are respectively $x + 2y - 3 = 0$, $2x - y + 4 = 0$ and their respective lengths are $\sqrt{2}$ and $2/\sqrt{3}$. The equation of the hyperbola is

$$1) \frac{2}{5}(2x - y + 4)^2 - \frac{3}{5}(x + 2y - 3)^2 = 1$$

$$2) \frac{2}{5}(2x - y + 3)^2 - \frac{3}{5}(x + 2y - 3)^2 = 1$$

$$3) 2(2x - y + 4)^2 - 3(x + 2y - 3)^2 = 1$$

$$4) 2(x + 2y - 3)^2 - 3(2x - y + 4)^2 = 1$$

4. If a variable line has its intercepts on the coordinate axes e, e' , where $\frac{e}{2}, \frac{e'}{2}$ are the eccentricities of a hyperbola and its conjugate hyperbola, then the line always touches the circle $x^2 + y^2 = r^2$, where $r =$

$$1) 1 \quad 2) 2 \quad 3) 3 \quad 4) 1/2$$

5. If e_1, e_2, e_3 are the eccentricities of a parabola (P) ellipse (E) and Hyperbola (H) respectively and $e_1^2 + e_2^2 + e_3^2 = \frac{46}{9}$, $e_1^2 - e_2^2 + e_3^2 = \frac{44}{9}$ then the eccentricity of conjugate hyperbola of H is

$$1) \sqrt{2} \quad 2) \frac{2}{\sqrt{3}} \quad 3) \frac{4}{3} \quad 4) 2$$

6. If a variable circle $x^2 + y^2 - 2ax + 4ay = 0$ intersects the hyperbola $xy = 4$ at the points (x_i, y_i) where $i = 1, 2, 3, 4$ then locus of the point $\left(\frac{x_1 + x_2 + x_3 + x_4}{4}, \frac{y_1 + y_2 + y_3 + y_4}{4} \right)$ is:

$$(1) y + 2x = 0 \quad (2) y - 2x + 5 = 0$$

$$(3) y - 2x = 0 \quad (4) y + 4x - 7 = 0$$

7. If two distinct tangents can be drawn from the point $(\alpha, 2)$ on different branches of the hyperbola $\frac{x^2}{9} - \frac{y^2}{16} = 1$, then

$$1. |\alpha| < \frac{3}{2} \quad 2. |\alpha| > \frac{2}{3} \quad 3. |\alpha| > 3 \quad 4. \alpha = 1$$

8. If $x = 9$ is the chord of contact of the Hyperbola $x^2 - y^2 = 9$ then the equation of the corresponding pair of tangents is

$$1) 9x^2 - 8y^2 + 18x - 9 = 0 \quad 2) 9x^2 - 8y^2 - 18x + 9 = 0$$

$$3) 9x^2 - 8y^2 - 18x - 9 = 0$$

$$4) 9x^2 - 8y^2 + 18x + 9 = 0$$

9. Consider a branch of the hyperbola $x^2 - 2y^2 - 2\sqrt{2}x - 4\sqrt{2}y - 6 = 0$ with vertex at the point A. Let B be one of the end points of its latus rectum. If C is the focus of the hyperbola nearest to the point A, then the area of the triangle ABC is [IIT : 2008]

1) $1 - \sqrt{\frac{2}{3}}$ 2) $\sqrt{\frac{3}{2}} - 1$ 3) $1 + \sqrt{\frac{2}{3}}$ 4) $\sqrt{\frac{3}{2}} + 1$

10. For hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$, let n be the number of points on the plane through which perpendicular tangents are drawn which of the following is not true

- 1) if $n = 1$, then $e = \sqrt{2}$ 2) if $n > 1$, then $0 < e < \sqrt{2}$
3) if $n = 0$, then $e > \sqrt{2}$ 4) if $n = 2$, then $e = 2$

11. PQ and RS are two perpendicular chords of the rectangular hyperbola $xy = c^2$. If C is the centre of the rectangular hyperbola, then the product of the slopes of CP, CQ, CR and CS is equal to

(1) -1 (2) 1 (3) 0 (4) ± 1

12. If two points P and Q on the hyperbola

$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$, which centre C be such that CP is

perpendicular to CQ, then $\frac{1}{CP^2} + \frac{1}{CQ^2}$ is equal to

1) $-\frac{1}{a^2} + \frac{1}{b^2}$ 2) $\frac{1}{a^2} + \frac{1}{b^2}$ 3) $-\frac{1}{a^2} - \frac{1}{b^2}$ 4) $\frac{1}{a^2} - \frac{1}{b^2}$

13. With one focus of the hyperbola $\frac{x^2}{9} - \frac{y^2}{16} = 1$

as the centre, a circle is drawn which is tangent to the hyperbola with no part of the circle being outside the hyperbola. The radius of the circle is

1) less than 2 2) 2 3) $1/3$ 4) 4

14. A ray emanating from the point (5, 0) is incident on the hyperbola $9x^2 - 16y^2 = 144$ at the point P with abscissa 8, then the equation of the reflected ray after first reflection is

1. $\sqrt{3}x - y + 7 = 0$
2. $3\sqrt{3}x - 13y + 15\sqrt{3} = 0$
3. $3\sqrt{3}x + 13y - 15\sqrt{3} = 0$ 4. $\sqrt{3}x + y - 14 = 0$

15. An ellipse $\frac{x^2}{4} + \frac{y^2}{3} = 1$ confocal with the hyperbola $\frac{x^2}{\cos^2 \theta} - \frac{y^2}{\sin^2 \theta} = 1$ then the set of values of ' θ '

1. R 2. $R - \{n\pi, n\pi + \pi\}$

3. $R - \left\{(2n+1)\frac{\pi}{2}, n\pi + \pi\right\}$ 4. $R - \left\{\frac{n\pi}{2}, n\pi + \pi\right\}$

16. If e_1 and e_2 are the eccentricities of the hyperbolas $xy = 9$ and $x^2 - y^2 = 25$, then (e_1, e_2) lie on a circle C_1 with centre origin then the (radius)² of the director circle of C_1 is

1) 2 2) 4 3) 8 4) 16

17. For a given non-zero value of m each of the lines $\frac{x}{a} - \frac{y}{b} = m$ and $\frac{x}{a} + \frac{y}{b} = m$ meets the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ at a point. Sum of the ordinates of these points, as m varies is

1) $\frac{a(1+m^2)}{m}$ 2) $\frac{b(1-m^2)}{m}$ 3) 0 4) $\frac{a+b}{2m}$

18. If radii of director circles of $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ and $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ are $2r$ and r respectively and e_e and e_h be the eccentricities of the ellipse and the hyperbola respectively then

1) $2e_h^2 - e_e^2 = 6$ 2) $e_e^2 - 4e_h^2 = 6$

3) $4e_h^2 - e_e^2 = 6$ 4) $e_e^2 - 2e_h^2 = 0$

19. If a hyperbola has one focus at the origin and its eccentricity is $\sqrt{2}$. One of the directrices is $x + y + 1 = 0$ then the equations to its asymptotes

1) $x = 1, y = 1$ 2) $x + 1 = 0, y + 1 = 0$

3) $x = 3, y = 3$ 4) $x + 3 = 0, y + 3 = 0$

20. A hyperbola, having the transverse axis of length $2\sin \theta$, is confocal with the ellipse $3x^2 + 4y^2 = 12$. Then its equation is

1) $x^2 \cos^2 \theta - y^2 \sec^2 \theta = 1$

2) $x^2 \sec^2 \theta - y^2 \cos^2 \theta = 1$

3) $x^2 \sin^2 \theta - y^2 \cos^2 \theta = 1$

4) $x^2 \cos^2 \theta - y^2 \sin^2 \theta = 1$

LEVEL - III - KEY

- 1) 1 2) 3 3) 1 4) 2 5) 2 6) 1 7) 1
 8) 2 9) 2 10) 4 11) 2 12) 4 13) 2 14) 2
 15) 4 16) 3 17) 3 18) 3 19) 2 20) 1

LEVEL - III - HINTS

1. Homogenisation and $a + b = 0 \therefore P = \frac{ab}{\sqrt{b^2 - a^2}}$
 The line touches the circle $\Rightarrow d = r \Rightarrow p = r$
2. Equation of tangent is $y = mx + \sqrt{9m^2 - 16}$
 $\therefore \sqrt{9m^2 - 16} = 2\sqrt{5} \Rightarrow m = \pm 2 \therefore a + b = 0$
3. Equation of the hyperbola is

$$\frac{\left(\frac{a_1x + b_1y + c_1}{\sqrt{a_1^2 + b_1^2}}\right)^2}{a^2} - \frac{\left(\frac{b_1x - a_1y + c_2}{\sqrt{b_1^2 + a_1^2}}\right)^2}{b^2} = 1$$
 where $a_1x + b_1y + c_1 = 0$, $b_1x - a_1y + c_2 = 0$ are conjugate and transverse axes respectively and a , b are lengths of semitransverse and semiconjugate axes respectively
4. Equation of the line is $\frac{x}{e} + \frac{y}{e'} = 1$

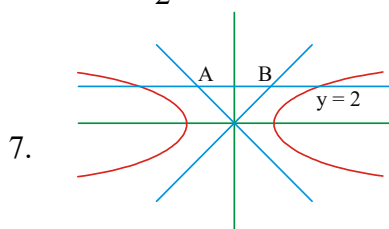
$$\frac{1}{\left(\frac{e}{2}\right)^2} + \frac{1}{\left(\frac{e'}{2}\right)^2} = 1 \Rightarrow \frac{1}{e^2} + \frac{1}{e'^2} = \frac{1}{4} \therefore d = r$$

$$\Rightarrow \frac{|-1|}{\sqrt{\frac{1}{e^2} + \frac{1}{e'^2}}} = r \Rightarrow r = 2$$
5. $e_1 = 1$ by adding two equations $\Rightarrow e_3 = 2$

$$\frac{1}{e_3^2} + \frac{1}{e_1^2} = 1$$
6. Putting $y = \frac{4}{x}$ in the equation of the circle, we get

$$x^4 - 2ax^3 + 16ax + 16 = 0 \text{ So } \frac{x_1 + x_2 + x_3 + x_4}{4} = \frac{a}{2}$$
 Similarly $\frac{y_1 + y_2 + y_3 + y_4}{4} = -a$

$$\Rightarrow h = \frac{a}{2}, k = -a \Rightarrow h = -\frac{k}{2} \Rightarrow y + 2x = 0$$



For two distinct tangents on different branches the point should lie on the line $y = 2$ and between A and B (where A and B are the points on the asymptotes). Equation of asymptotes are

$$4x = \pm 3y. \text{ Solving with } y = 2, \text{ we have } x = \pm \frac{3}{2}$$

$$-\frac{3}{2} < \alpha < \frac{3}{2}$$

8. Equation of chord of contact of $P(x_1, y_1)$ is $S_1 = 0$ compare with $x = 9$, find $P(x_1, y_1)$. Equation of pair of tangents from $P(x_1, y_1)$ is $S_1^2 = S.S_{11}$
9. Given Equation can be written as

$$\frac{(x - \sqrt{2})^2}{4} - \frac{(y + \sqrt{2})^2}{2} = 1$$

$$a^2 = 4, b^2 = 2; a = 2, b = \sqrt{2}; e = \sqrt{\frac{3}{2}}$$
 area of $\Delta ABC = \frac{1}{2}(AC)(BC) = \frac{1}{2}(ae - a) \cdot \frac{b^2}{a}$

$$= \frac{b^2}{2}(e - 1) = \frac{2}{2}\left(\sqrt{\frac{3}{2}} - 1\right) = \sqrt{\frac{3}{2}} - 1$$
10. Locus of point of intersection of perpendicular tangents is director circle $x^2 + y^2 = a^2 - b^2$. Now,
 $e^2 = 1 + \frac{b^2}{a^2}$. If $a^2 > b^2$, then there are infinite (or more than 1) points on the circle $\Rightarrow e^2 < 2 \Rightarrow e < \sqrt{2}$. If $a^2 < b^2$, there does not exist any point on the plane $\Rightarrow e^2 > 2 \Rightarrow e > \sqrt{2}$. If $a^2 = b^2$, there is exactly one point (centre of the hyperbola) $\Rightarrow e = \sqrt{2}$
11. let t_1, t_2, t_3, t_4 be the parameters of the points P, Q, R, S respectively.
 $\therefore P$ is $\left(ct_1, \frac{c}{t_1}\right)$, Q is $\left(ct_2, \frac{c}{t_2}\right)$, R is $\left(ct_3, \frac{c}{t_3}\right)$, S is $\left(ct_4, \frac{c}{t_4}\right)$ Since $PQ \perp RS$

12. Let $CP = r_1$ be inclined to transverse axis at an angle θ so that P is $(r_1 \cos \theta, r_1 \sin \theta)$ and P lies on the hyperbola $\Rightarrow r_1^2 \left(\frac{\cos^2 \theta}{a^2} - \frac{\sin^2 \theta}{b^2} \right) = 1$

Replacing θ by $90^\circ + \theta$, then $r_2^2 \left(\frac{\sin^2 \theta}{a^2} - \frac{\cos^2 \theta}{b^2} \right) = 1$

$$\Rightarrow \frac{1}{r_1^2} + \frac{1}{r_2^2} = \frac{\cos^2 \theta}{a^2} - \frac{\sin^2 \theta}{b^2} + \frac{\sin^2 \theta}{a^2} - \frac{\cos^2 \theta}{b^2}$$

$$\Rightarrow \frac{1}{r_1^2} + \frac{1}{r_2^2} = \cos^2 \theta \left(\frac{1}{a^2} - \frac{1}{b^2} \right) + \sin^2 \theta \left(\frac{1}{a^2} - \frac{1}{b^2} \right)$$

$$\Rightarrow \frac{1}{r_1^2} + \frac{1}{r_2^2} = \frac{1}{a^2} - \frac{1}{b^2} \Rightarrow \frac{1}{CP^2} + \frac{1}{CQ^2} = \frac{1}{a^2} - \frac{1}{b^2}$$

13. Focus $(ae, 0) = (5, 0)$

Equation of circle is $(x-5)^2 + y^2 = r^2$ —(1)

Equation of hyperbola is $\frac{x^2}{9} - \frac{y^2}{16} = 1$ —(2)

(1), (2) $\Rightarrow \frac{(x-5)^2}{16} + \frac{x^2}{9} = \frac{r^2}{16} + 1$

$25x^2 - 90x + 81 - 9r^2 = 0$

$x = \frac{9 \pm \sqrt{90^2 - 4(25)(81 - 9r^2)}}{50}$

$= \frac{9 \pm \sqrt{81 - 81 + 9r^2}}{5} = \frac{9 \pm 3r}{5} \geq 3$

Vertex of hyperbola is $(3, 0)$

$9 \pm 3r \geq 15 ; \pm 3r \geq 6 ; r \geq 2$

$r \geq 2$, circle touching are intersecting

14. Foci $(\pm 5, 0)$; $x = 8 \Rightarrow y = \pm 3\sqrt{3}$

Equation of the required line is the line joining

$S^1(-5, 0)$ and $P(8, \pm 3\sqrt{3})$

15. $e_1 = \sqrt{\frac{a^2 - b^2}{a^2}} = \frac{1}{2}$ foci $(\pm ae, 0) = (\pm 1, 0)$

$e_2 = \sqrt{\frac{a^2 + b^2}{a^2}} = \frac{1}{\cos \theta}$; $\cos \theta \neq 0$

16. $e_1 = \sqrt{2}, e_2 = \sqrt{2}$ equation of the circle C_1 is (e_1, e_2) lies on $C_1 \Rightarrow r^2 = 4$. equation of director circle of C_1 is $x^2 + y^2 = 2r^2$

17. Ordinate of the point of intersection of the line

$\frac{x}{a} - \frac{y}{b} = m$ and the hyperbola is given by

$\left(\frac{x}{a} - \frac{y}{b} \right) \left(\frac{x}{a} - \frac{y}{b} + \frac{2y}{b} \right) = 1$

i.e. $m \left(m + \frac{2y}{b} \right) = 1$ i.e. $y = \frac{b(1 - m^2)}{2m}$

Similarly ordinate of the point of intersection of the

line $\frac{x}{a} + \frac{y}{b} = m$ and the hyperbola is given by

$y = \frac{b(m^2 - 1)}{2m}$ $\therefore$ Sum of the ordinates is 0.

18. Equation of director circles of ellipse and hyperbola are respectively.

$x^2 + y^2 = a^2 + b^2$ and $x^2 + y^2 = a^2 - b^2$

$a^2 + b^2 = 4r^2$... (1) $a^2 - b^2 = r^2$... (2)

$a^2 = \frac{5r^2}{2}, b^2 = \frac{3r^2}{2}, e_c^2 = 1 - \frac{b^2}{a^2}$

$\Rightarrow e_c^2 = 1 - \frac{3r^2}{2} \times \frac{2}{5r^2} = 1 - \frac{3}{5} = \frac{2}{5}$; $e_h^2 = 1 + \frac{b^2}{a^2}$

$\Rightarrow e_h^2 = 1 + \frac{3}{5} = \frac{8}{5}$ So, $4e_h^2 - e_c^2 = 4 \times \frac{8}{5} - \frac{2}{5} = \frac{30}{5} = 6$

19. $sp^2 = e^2 pm^2 \Rightarrow x^2 + y^2 = (x + y + 1)^2$

$\Rightarrow 2xy + 2x + 2y + 1 = 0 \Rightarrow (x + 1)(y + 1) = 1$

Pair of equation of asymptotes $(x + 1)(y + 1) = 0$

20. The given ellipse is $\frac{x^2}{4} + \frac{y^2}{3} = 1$

$\Rightarrow a = 2, b = \sqrt{3} \Rightarrow 3 = 4(1 - e^2) \Rightarrow e = \frac{1}{2}$

so that $ae = 1$ Hence the eccentricity e_1 , of the hyperbola is given by $1 = e_1 \sin \theta \Rightarrow e_1 = \sec \theta$

$\Rightarrow b^2 = \sin^2 \theta (\sec^2 \theta - 1) = \cos^2 \theta$

Hence the hyperbola is $\frac{x^2}{\sin^2 \theta} - \frac{y^2}{\cos^2 \theta} = 1$ or

$x^2 \sec^2 \theta - y^2 \sec^2 \theta = 1$



LEVEL-IV

1. Match the following:

Equations of the curve Nature of the curve

A. $x = 2(\cos t + \sin t), y = 5(\cos t - \sin t)$

1. Parabola

B. $x = 3(\cosh \theta + \sinh \theta), y = 4(\cosh \theta - \sinh \theta)$

2. Ellipse

C. $x = \sin^2 t, y = 2 \cos t$ 3. Hyperbola

4. Rectangular hyperbola

The correct match is

1) A-1, B-2, C-3 2) A-2, B-3, C-1

3) A-3, B-2, C-1 4) A-2, B-4, C-1

2. Observe the following lists.

HYPERBOLA ECCENTRICITY

A. $x^2 - y^2 = 9$ 1. 1

B. $xy = 5$ 2. $\sqrt{2}$

C. $2x^2 - 3y^2 = 6$ 3. $\sqrt{\frac{5}{3}}$

D. $4(x-1)^2 - 2(y+2)^2 = 10$ 4. $\sqrt{3}$

The correct match is

	A	B	C	D
1)	1	2	3	4
2)	2	1	3	4
3)	2	2	3	4
4)	2	4	1	3

3. Observe the following list

HYPERBOLA

CENTRE

A. $2x^2 + 5xy + 2y^2 - 11x - 7y - 4 = 0$ 1. $\left(\frac{84}{25}, \frac{-12}{25}\right)$

B. $2xy + 3x + 4y + 1 = 0$ 2. $(-1, 3)$

C. $\frac{(3x - 4y - 12)^2}{225} - \frac{(4x + 3y - 12)^2}{100} = 1$ 3. $(6, 2)$

D. $72(x - 6)^2 - 9(y - 2)^2 = 32$ 4. $(3, -1)$

5. $\left(-2, \frac{-3}{2}\right)$

The correct match is

	A	B	C	D
1)	1	3	2	5
2)	2	5	1	3
3)	2	5	3	1
4)	1	3	5	2

4. Match the following:

Hyperbola

Auxiliary circle

I. $5x^2 - 9y^2 = 45$ a. $x^2 + y^2 = 2$

II. $2x^2 - 5y^2 + 10 = 0$ b. $x^2 + y^2 = 9$

III. $x^2 - 4y^2 + 16 = 0$ c. $x^2 + y^2 = 16$

IV. $xy = 8$ d. $x^2 + y^2 = 4$

e. $x^2 + y^2 = 8$

1) b,a,d,c 2) b,a,d,e 3) a,b,c,d 4) b,c,d,e

5. Match the following

Hyperbola

Length of Latusrectum

I. $x^2 - 4y^2 = 4$ a. 1

II. $25y^2 - 16x^2 = 400$ b. 12

III. $2x^2 - y^2 - 4x - 4y - 20 = 0$ c. $\frac{9}{2}$

IV. $9x^2 - 16y^2 + 72x - 32y - 16 = 0$ d. $\frac{25}{2}$

The correct match is

	I	II	III	IV
1.	a	b	c	d
2.	a	d	b	c
3.	c	a	b	d
4.	b	c	d	a

Assertion & Reason Type questions

1) Both A and R are true and R is the correct explanation of A.

2) Both A and R are true but R is not the correct explanation of A.

3) A is true but R is false. 4) A is false but R is true.

6. Assertion(A): The difference of the focal distances of any point on the hyperbola

$$\frac{x^2}{36} - \frac{y^2}{9} = 1 \text{ is } 12.$$

Reason(R): The difference of the focal distances of any point on the hyperbola is equal to the length of its transverse axis.

7. Assertion(A): The equation $(x + 1)(y - 3) = 4$ represents a rectangular hyperbola.

Reason(R): The equation

$$ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$$

represents a rectangular hyperbola if .

$$a+b=0, h^2 > ab \text{ and } abc+2fgh-af^2-bg^2-ch^2 \neq 0$$

8. Assertion(A): The product of the perpendiculars from the foci to any tangent to the

$$\text{hyperbola } \frac{x^2}{12} - \frac{y^2}{9} = 1 \text{ is } 9.$$

Reason(R): The product of the perpendiculars from the foci on any tangent to the hyperbola is square of the semi transverse axis.

9. Assertion (A): The number of tangents to the

$$\text{hyperbola } \frac{x^2}{16} - \frac{y^2}{9} = 1 \text{ having slope } \frac{1}{2} \text{ are zero}$$

Reason (R): If the slope of a tangent to

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1 \text{ is } m \text{ then } m \in \left(-\infty, \frac{-b}{a}\right] \cup \left[\frac{b}{a}, \infty\right)$$

10. I: The orthocentre of the triangle formed by any three points on $xy = c^2$ lies on the same hyperbola.

II: The triangle formed by any three points on $xy = c^2$ ($x > 0, y > 0$) is an obtuse angled triangle.

Which of the above statement is correct?

1) Only I

2) Only II

3) Both I and II

4) Neither I nor II

11. **Statement - 1 :** The point (7, -3) lies inside the hyperbola $9x^2 - 4y^2 = 36$ where as the point (2, 7) lies outside this.

Statement - 2 : The point (x_1, y_1) lies outside,

on or inside the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$

according as $\frac{x^2}{a^2} - \frac{y^2}{b^2} - 1 < 0$ or $= 0$ or > 0

Which of the above statement is correct?

- 1) Only I 2) Only II
3) Both I and II 4) Neither I nor II

12. **Arrange the eccentricities of the following hyperbola in the descending order**

A. In a hyperbola, the distance between the foci is three times the distance between the directrices

B. In a hyperbola, the transverse axis is double the conjugate axis

C. $axy = b + c$ ($abc > 0$)

- 1) A, C, B 2) B, C, A 3) A, B, C 4) C, A, B

13. **The arrangement of the following hyperbolas in the ascending order of the radius of their director circles.**

A) $16x^2 - 25y^2 = 400$

B) $4x^2 - 9y^2 = 36$

C) $9x^2 - 16y^2 = 144$

D) $3x^2 - 4y^2 = 12$

- 1) A, B, C, D 2) A, C, B, D
3) D, B, C, A 4) D, C, B, A

LEVEL - IV - KEY

- 1) 4 2) 3 3) 2 4) 1 5) 2 6) 1 7) 1
8) 3 9) 1 10) 3 11) 3 12) 1 13) 3

LEVEL - IV - HINTS

1. a) $\frac{x^2}{4} + \frac{y^2}{25} = 2$ b) $xy = 12$ c) $y^2 = 4(1-x)$
2. a) $x^2 - y^2 = 9$ is rectangular hyperbola
b) $xy = 5 \Rightarrow e = \sqrt{2}$
c) $\frac{x^2}{3} - \frac{y^2}{2} = 1 \Rightarrow e = \sqrt{\frac{3+2}{2}} = \sqrt{\frac{5}{2}}$

d) $\frac{(x-1)^2}{5/2} - \frac{(y+2)^2}{5} = 1 \Rightarrow e = \sqrt{\frac{\frac{5}{2}+5}{\frac{5}{2}}} = \sqrt{3}$

3. Solve $\frac{\partial s}{\partial x} = 0, \frac{\partial s}{\partial y} = 0$

4. Auxiliary circle of $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ is $x^2 + y^2 = a^2$

Auxiliary circle of $\frac{x^2}{a^2} - \frac{y^2}{b^2} = -1$ is $x^2 + y^2 = b^2$

5. Length of the latus rectum = $\left\{ \begin{array}{l} \frac{2b^2}{a} \text{ for } \frac{x^2}{a^2} - \frac{y^2}{b^2} = 1 \\ \frac{2a^2}{b} \text{ for } \frac{x^2}{a^2} - \frac{y^2}{b^2} = -1 \end{array} \right\}$

6. $|SP - S'P| = 2a$

7. The equation $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$ represents a rectangular hyperbola if $a+b=0$, $h^2 > ab$, $abc + 2fgh - af^2 - bg^2 - ch^2 \neq 0$

8. The product of the perpendiculars from the foci to any tangent to the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ is b^2

9. Using $c^2 = a^2 m^2 - b^2$

10. I. Orthocentre = $\left(\frac{-c}{t_1 t_2 t_3}, -ct_1 t_2 t_3 \right)$

II. The triangle formed by any three points $xy = c^2$ ($x > 0, y > 0$) is an obtuse angled triangle.

11. The point $P(x_1, y_1)$ lies inside the hyperbola $S = 0$. If $S_{11} > 0$, lies outside if $S_{11} < 0$ and on the hyperbola if $S_{11} = 0$.

12. a) $2ae = 3 \left(2\frac{a}{e} \right) \Rightarrow e = \sqrt{3}$

b) $2a = 3(2b) \Rightarrow e = \sqrt{\frac{5}{4}}$ c) $e = \sqrt{2}$

13. Equation of the director circle of $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ is $x^2 + y^2 = a^2 - b^2$